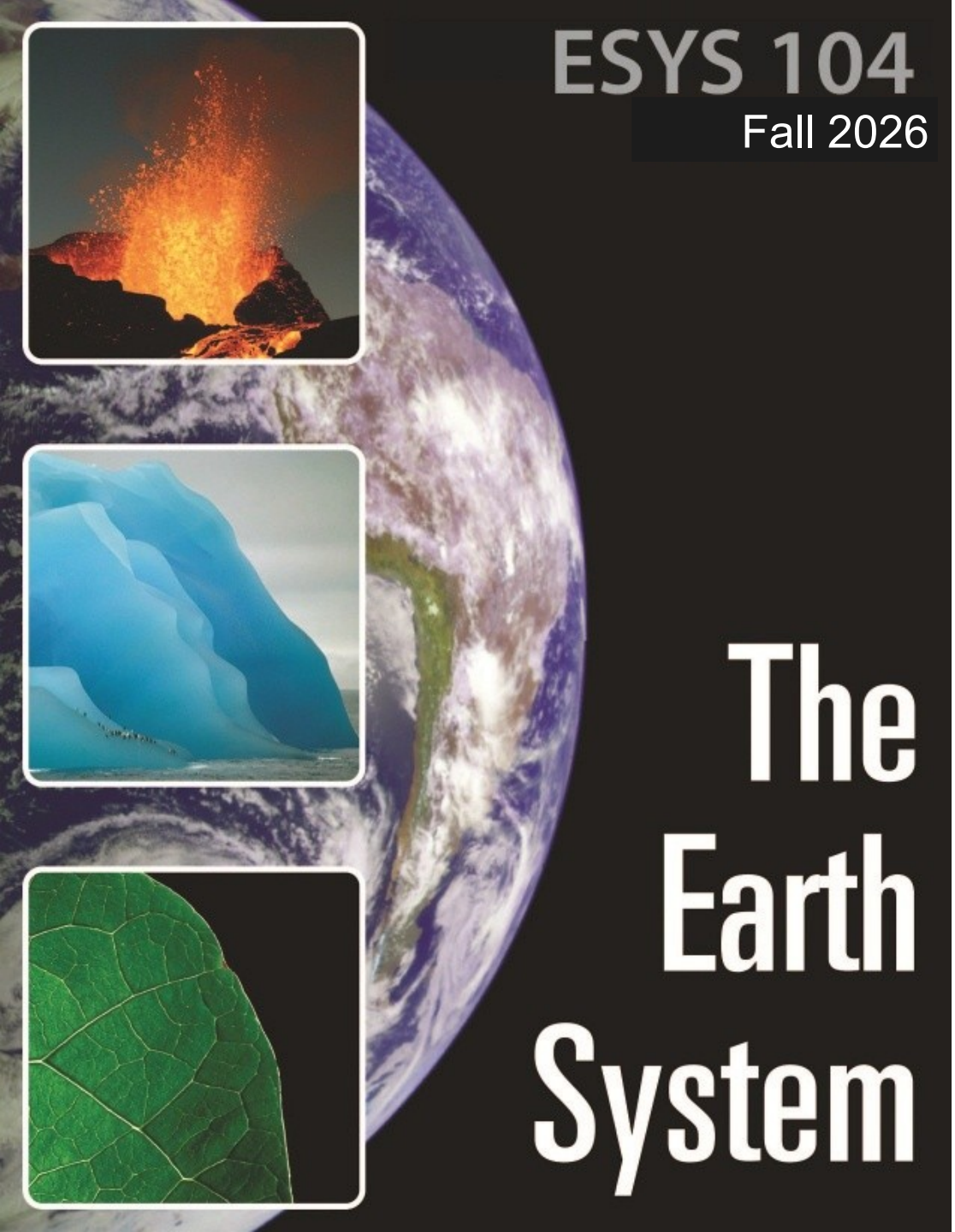


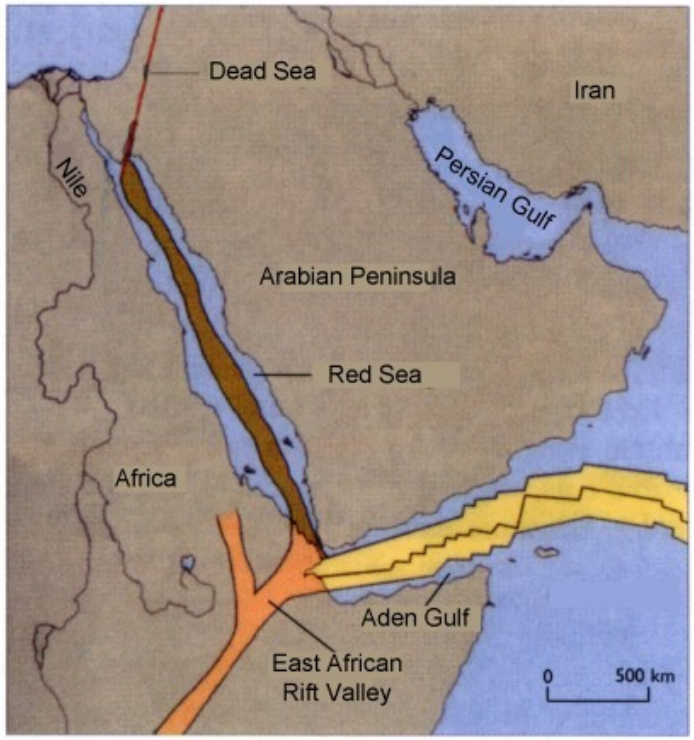
#4: Resources and Reservoirs

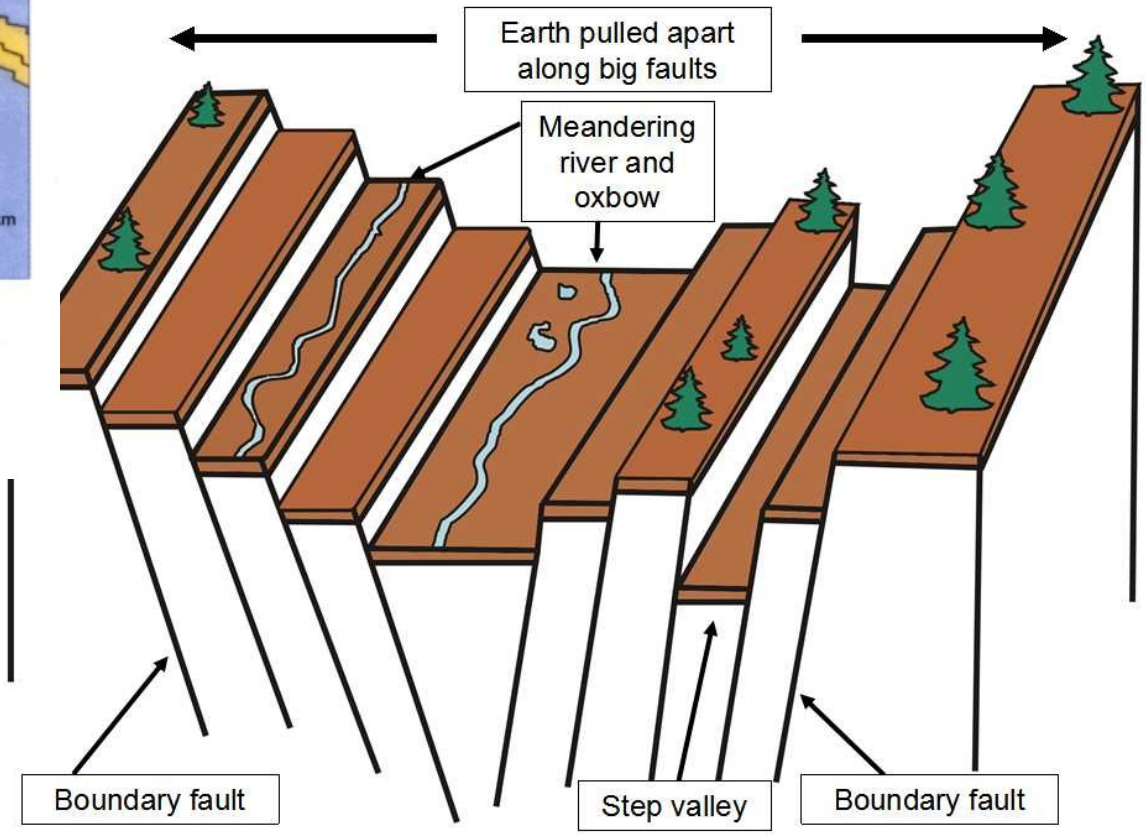
Atmospheres on Wednesday (new readings)

Sample Quiz available on myCourses until midnight tonight


THE FACULTY OF SCIENCE PRESENTS - THE 42ND EDITION OF
SOUP AND SCIENCE
SEPTEMBER 14 - 18, 2026
THOMSON HOUSE - BALLROOM
11:45 - 12:45
LUNCH PROVIDED
FIRST COME, FIRST SERVED
N_2O
CH_4
CO_2
A G C T
10011001
11010100
00010011
$E=mc^2$
$PV=nRT$
Ψ
$\pm$
$\mp$

East African Rift



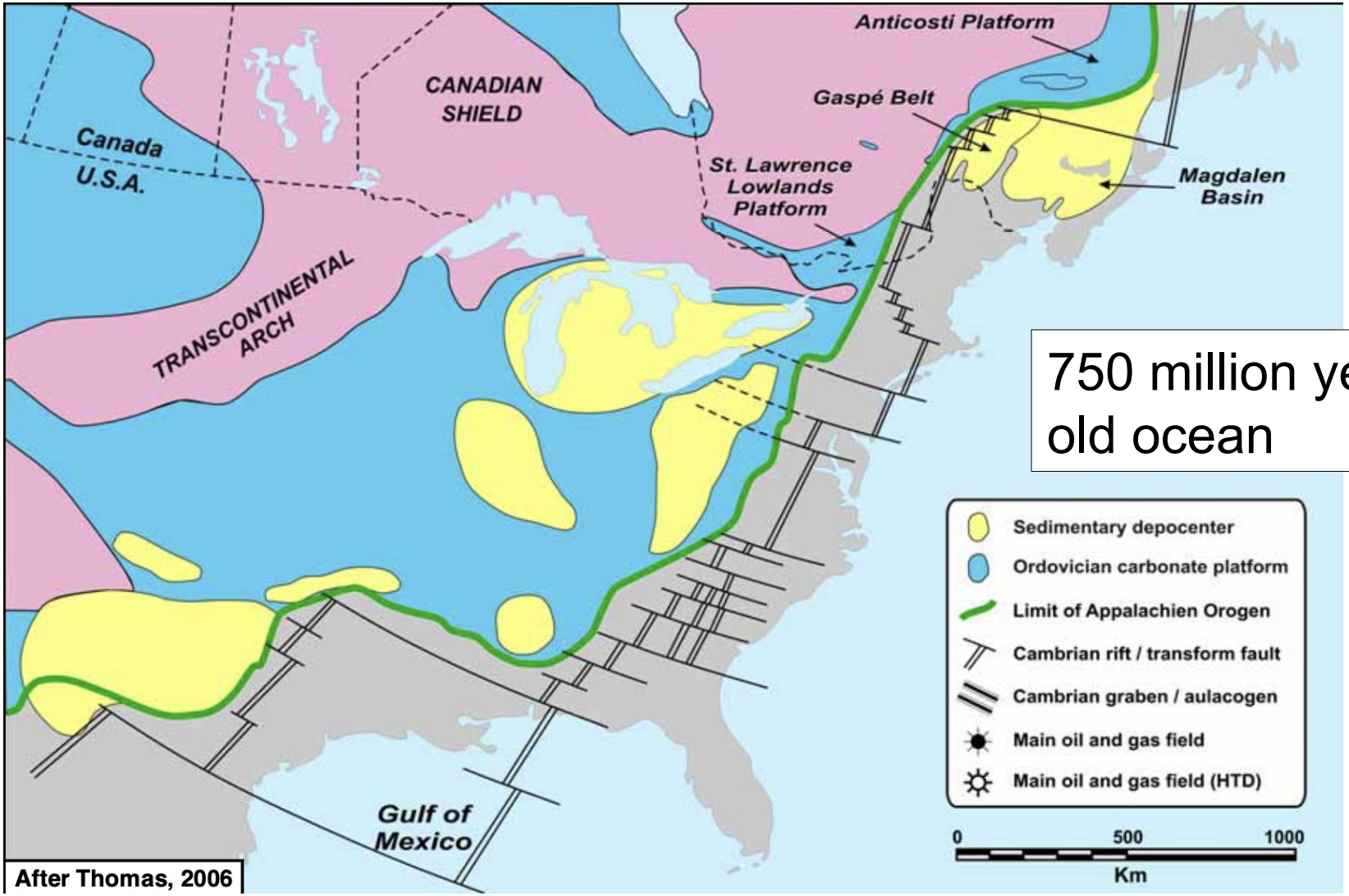


Before Pangea: the Supercontinent Rodinia

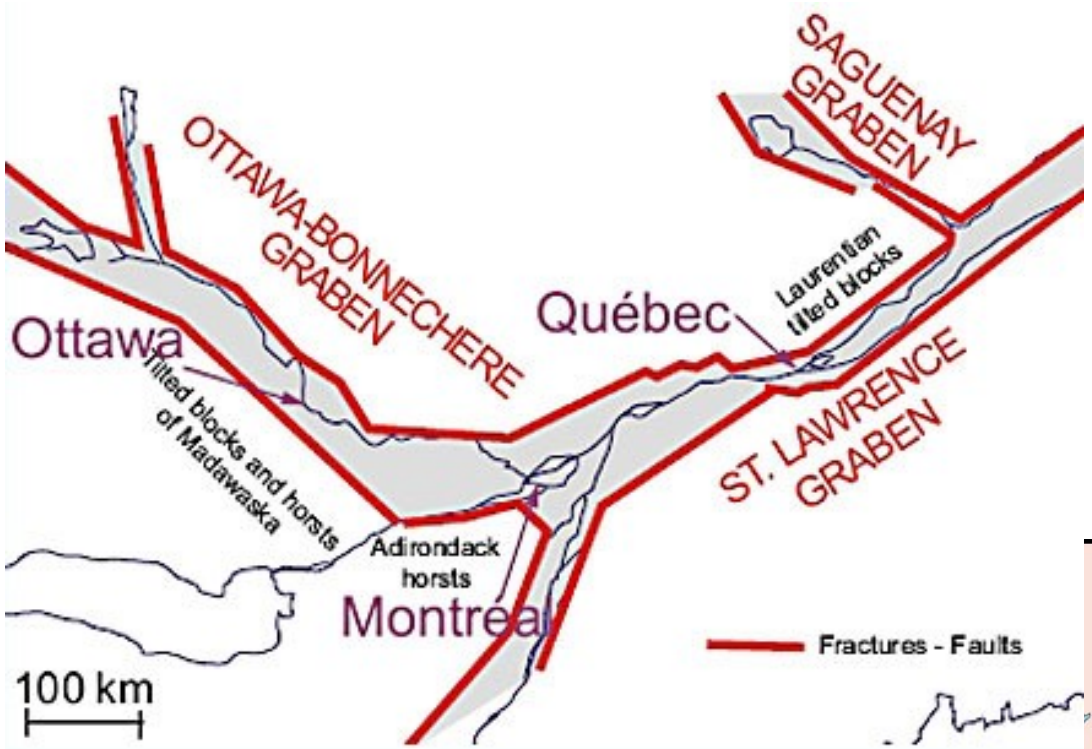
Rodinia
1.2 billion years ago



Ottawa and St. Lawrence valleys: old rifts



Ottawa and St. Lawrence valleys: old rifts



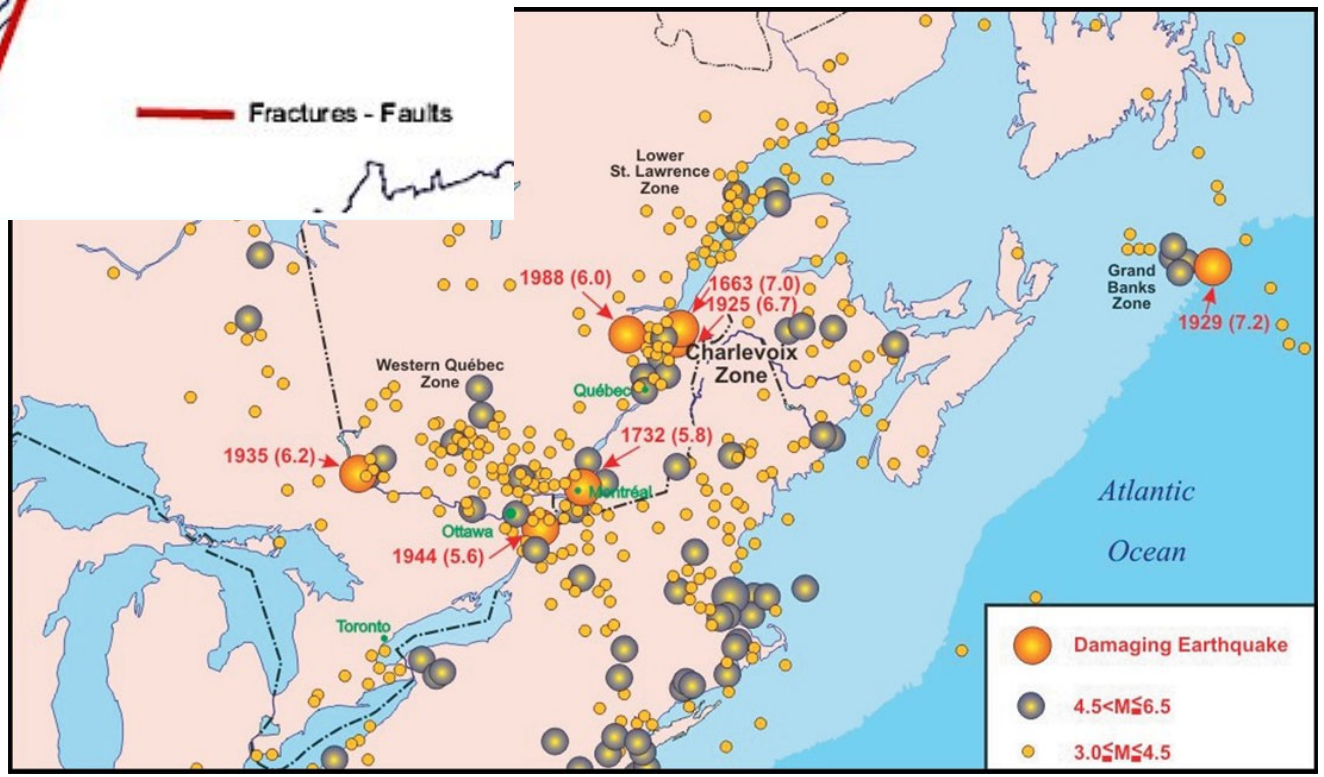
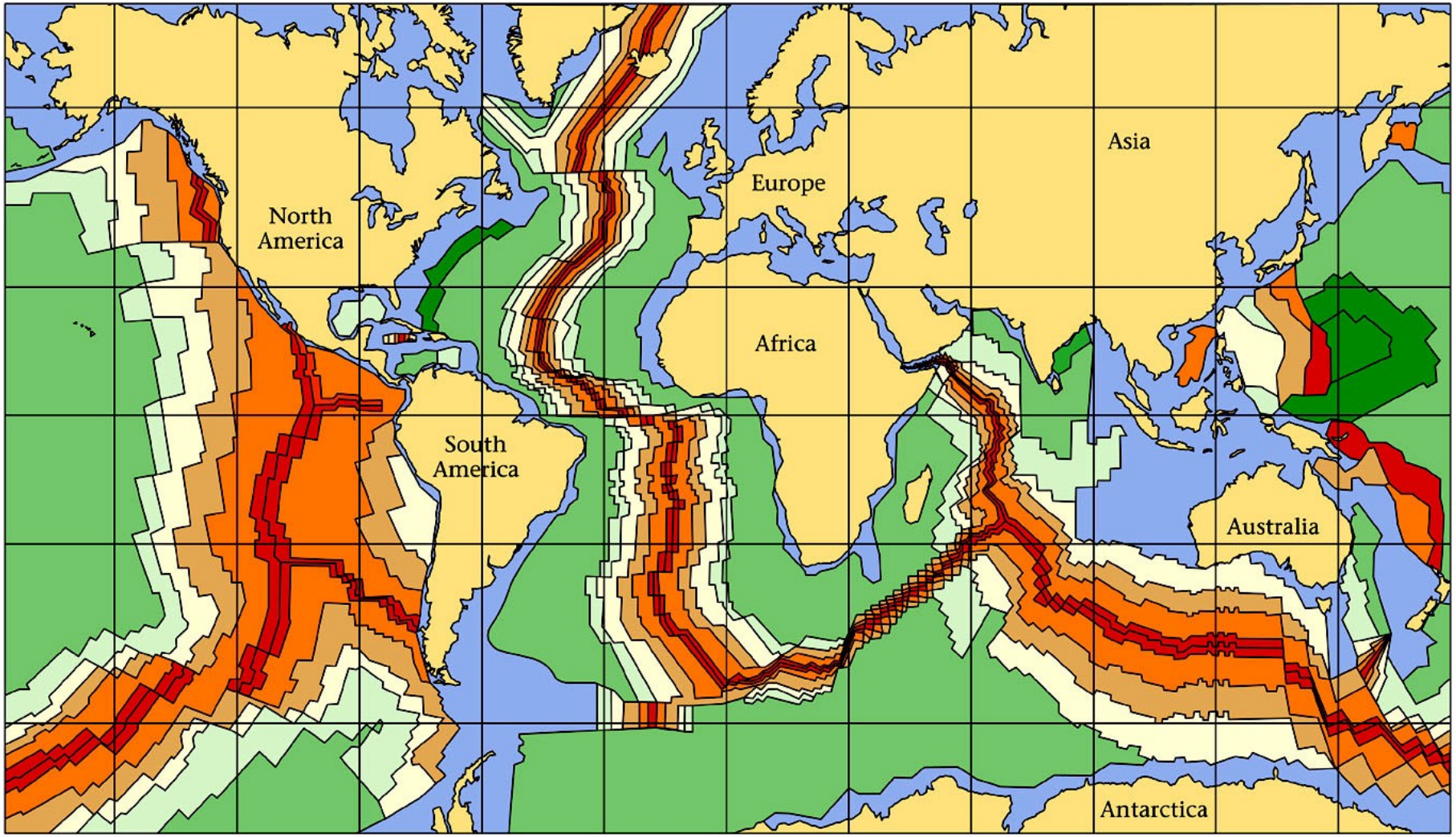
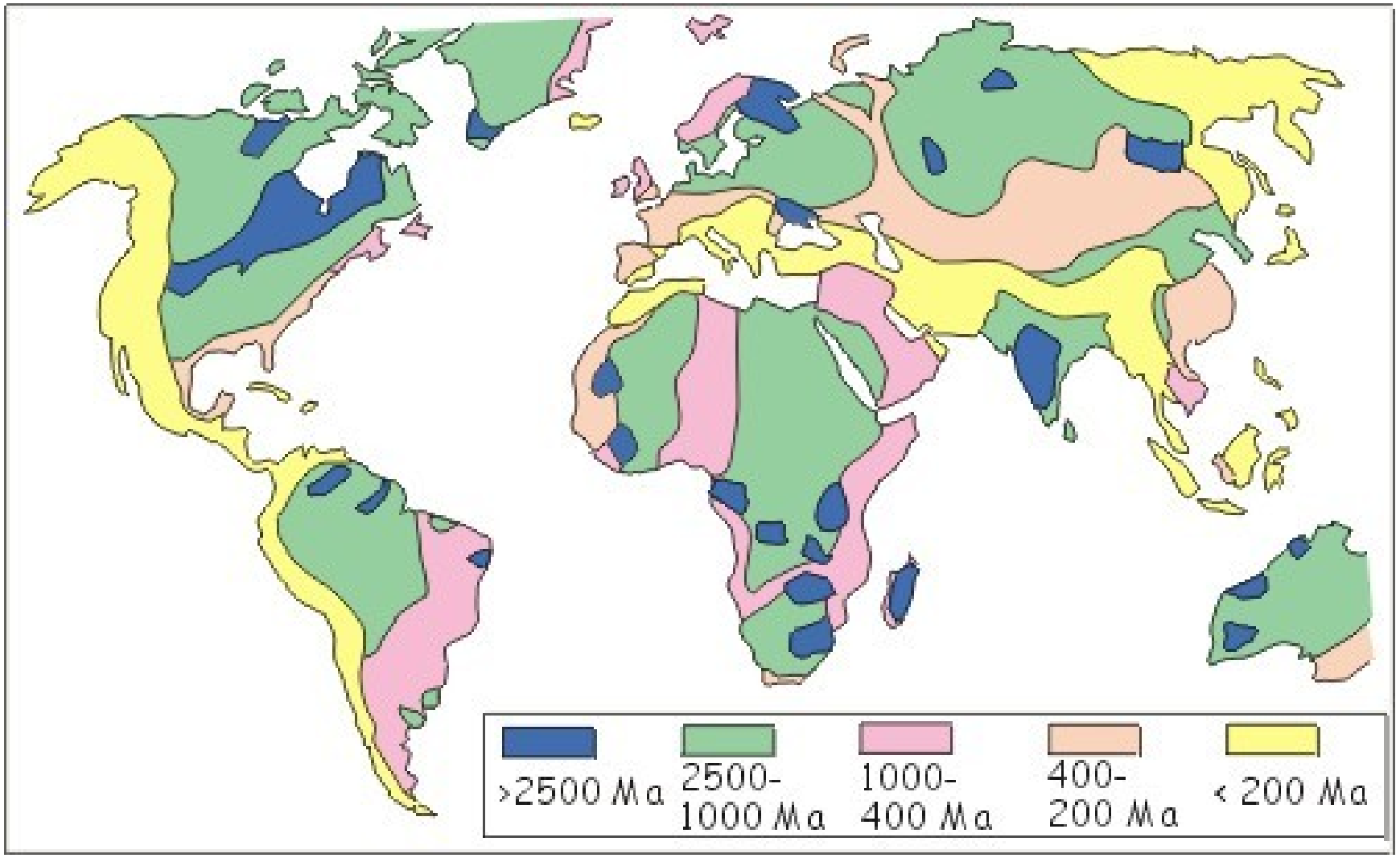


Plate Tectonics: Ocean Crust Age



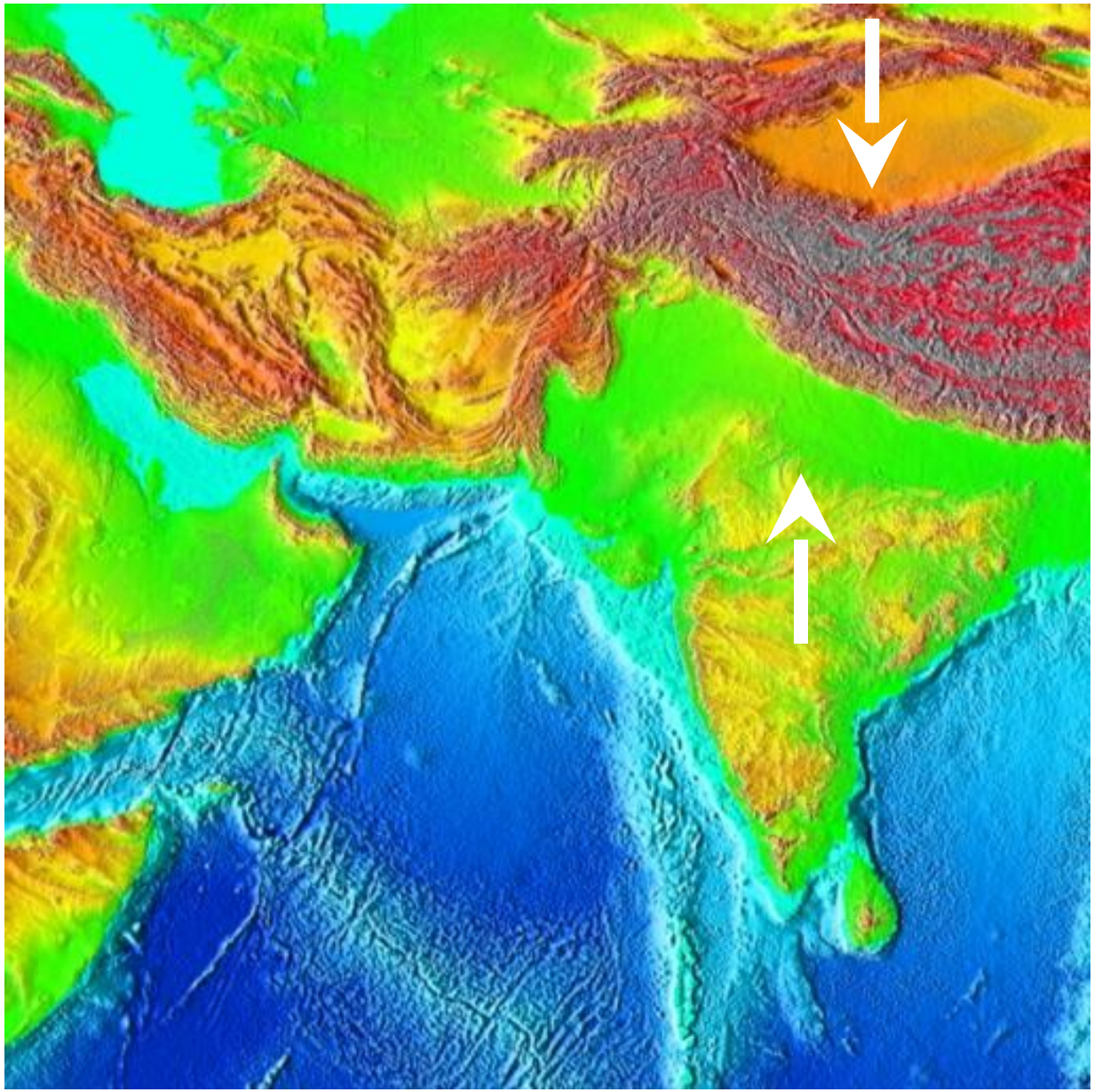
Ma	0–5	5–21	21–38	38–52	52–65	65–145	145–160
Age	Pleistocene to Pliocene	Miocene	Oligocene	Eocene	Paleocene	Cretaceous	Jurassic

Plate Tectonics: Continental Crust Age



Ma = millions of years

Convergent Margins – The Himalayas



Plates: Venus vs. Earth

Both Earth and Venus have similar concentrations of carbon and oxygen, but their atmospheres and surfaces are very different

The Greenhouse effect!

At the surface

Venus	Earth
95 bars pressure	1 bar pressure
96 % CO_2	78 % N_2
737 K (601 K lead melts)	288 K (273 K water ice melts)

But if Venus and Earth have the same amount of carbon and oxygen – where is Earth's CO_2?

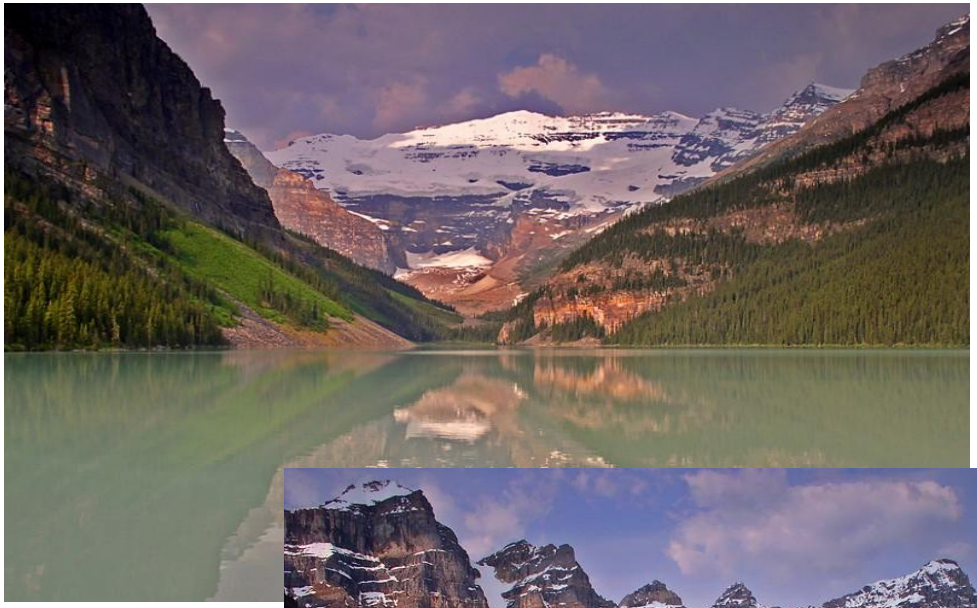


In calcite, $CaCO_3$, in the rock limestone.

This is atomic scale control on Earth's atmosphere

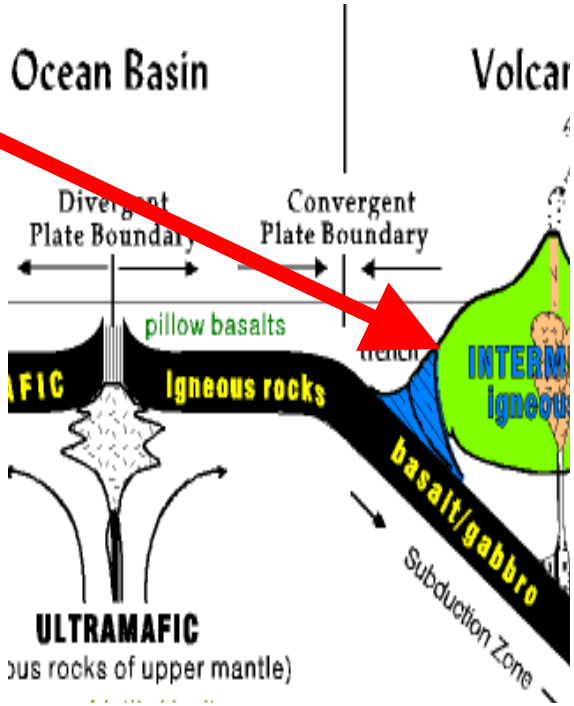
And where is Earth's limestone?

In mountain ranges or continental margins





Banff National Park

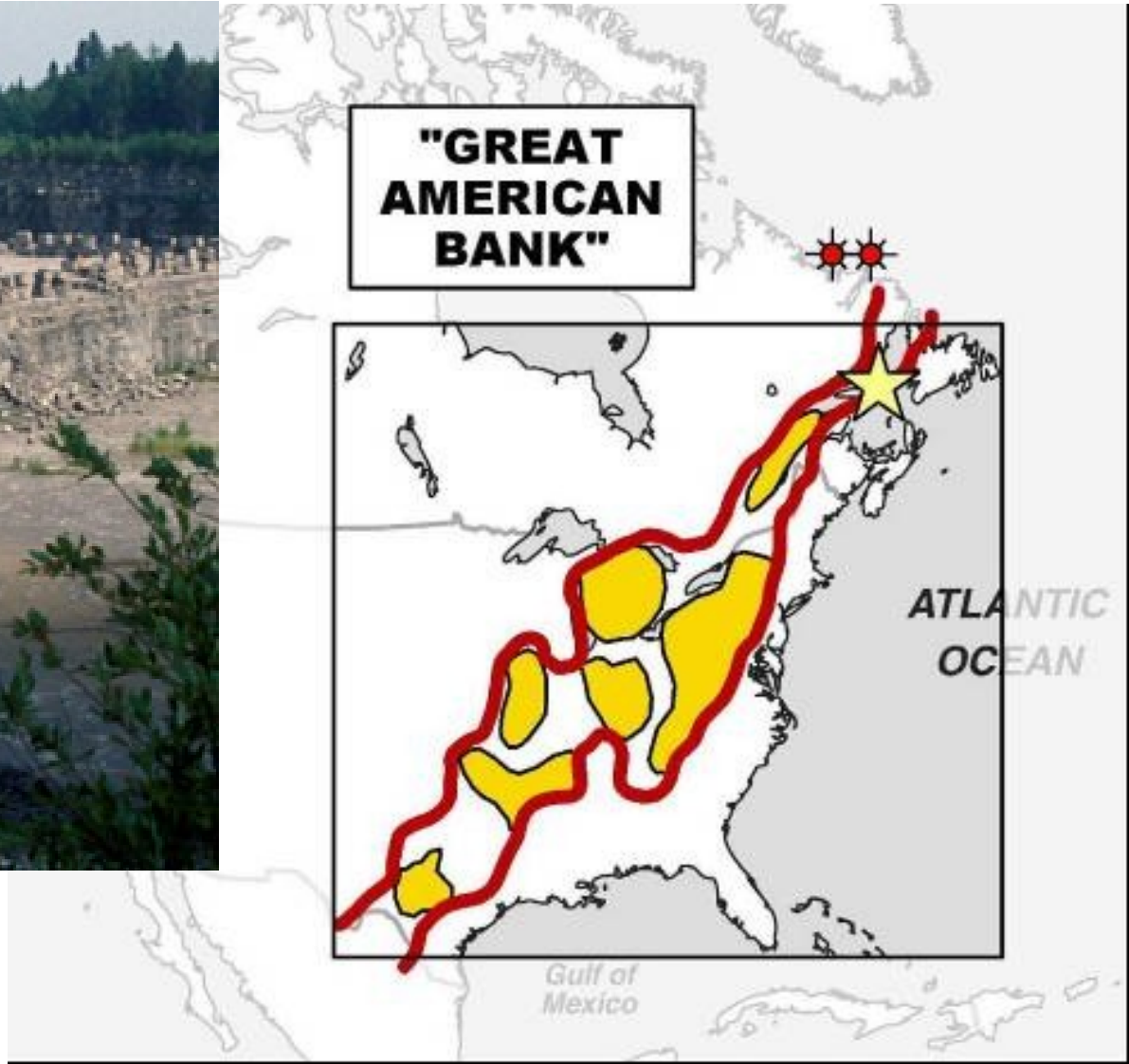


http://www.pbase.com/turnstyle/landscapes

And where is Earth's limestone?

In sediment on continents deposited in shallow seas (the St. Laurence lowlands including Montreal)





Chemical Weathering: Hydrolysis

Weak acids strip cations out of minerals

These reactions requires water to be Present: either as liquid or as vapor

$CaAl_2Si_2O_8$ + $2H_2CO_3$ + H_2O →

feldspar + carbonic acid + water

$Al_2Si_2O_5(OH)_4$ + Ca^{++} + $2\ HCO_3^-$

kaolinite (a clay) + calcium (dissolved) + bicarbonate (dissolved)

Organisms use dissolved calcium and bicarbonate to produce calcite shells.

CO_2, rocks and the atmosphere

So exposing silicate rocks to the atmosphere can cause them to react and remove CO_2 from the atmosphere

Simplified weathering reactions (water and acids not shown)

$$CaSiO_3 + CO_2 \Rightarrow CaCO_3 + SiO_2$$

But, metamorphosing or weathering carbonates releases their CO_2 into the atmosphere.

$$CaCO_3 + SiO_2 \Rightarrow CaSiO_3 + CO_2$$

The relative rates of these processes are important

How long is CO_2 stored?--The Residence Time

The average duration an element or molecule is stored in a reservoir at steady state conditions is named the "residence time". This average time is calculated by dividing the mass of the element or molecule by either the flux into, or out of, the reservoir (they're the same at steady state). Or using math:

$$\text{Residence time} = \frac{m}{dm/dt}$$

In continental rocks the residence time of carbon is ~ 150 Ma.

On the other hand carbon in soils (contain both calcite and organic carbon) has a residence time of about 25 years.

Atmospheric Carbon Residence Time

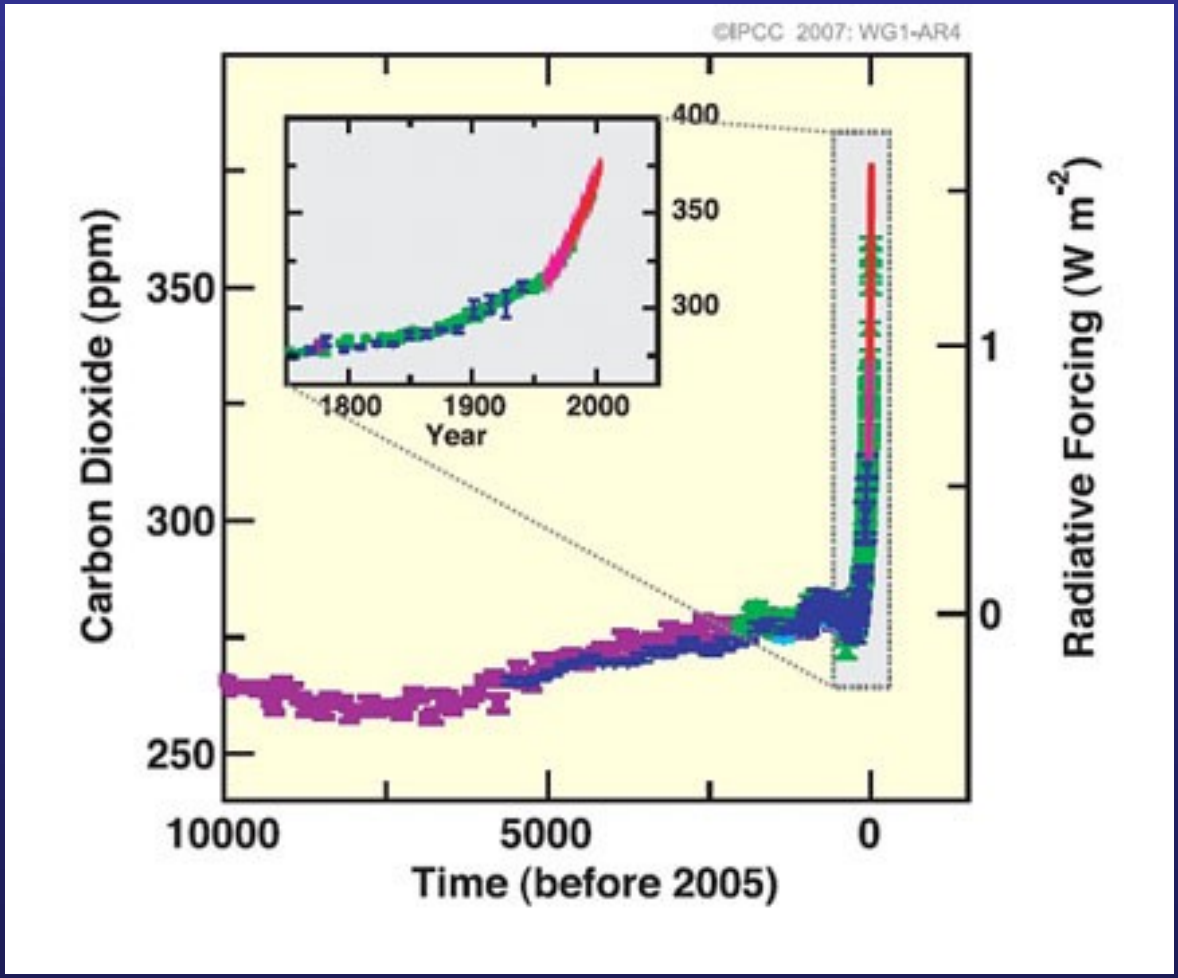
Global Emissions of Carbon: 10 Gigatonnes/year
(40 Gigatonnes CO_2)

Atmospheric Reservoir: 750 Gigatonnes

Residence time: 70 - 80 years

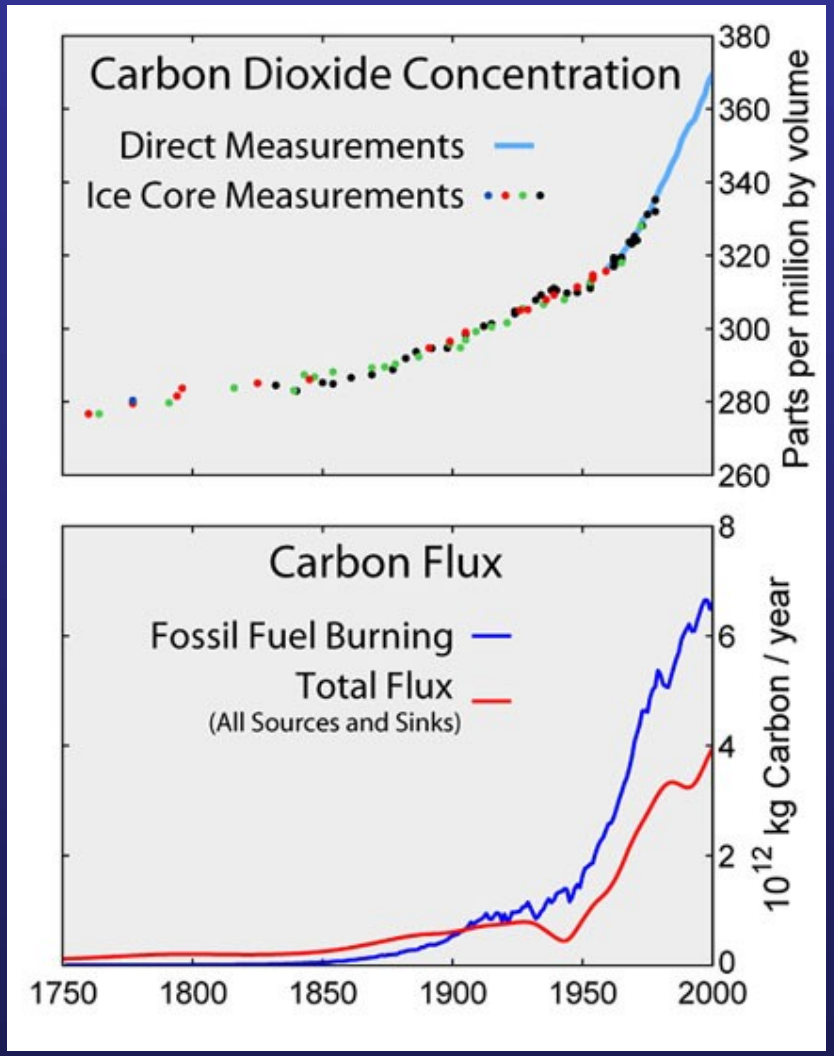
Ocean Reservoir: 38,000 Gigatonnes

Atmospheric Carbon Dioxide



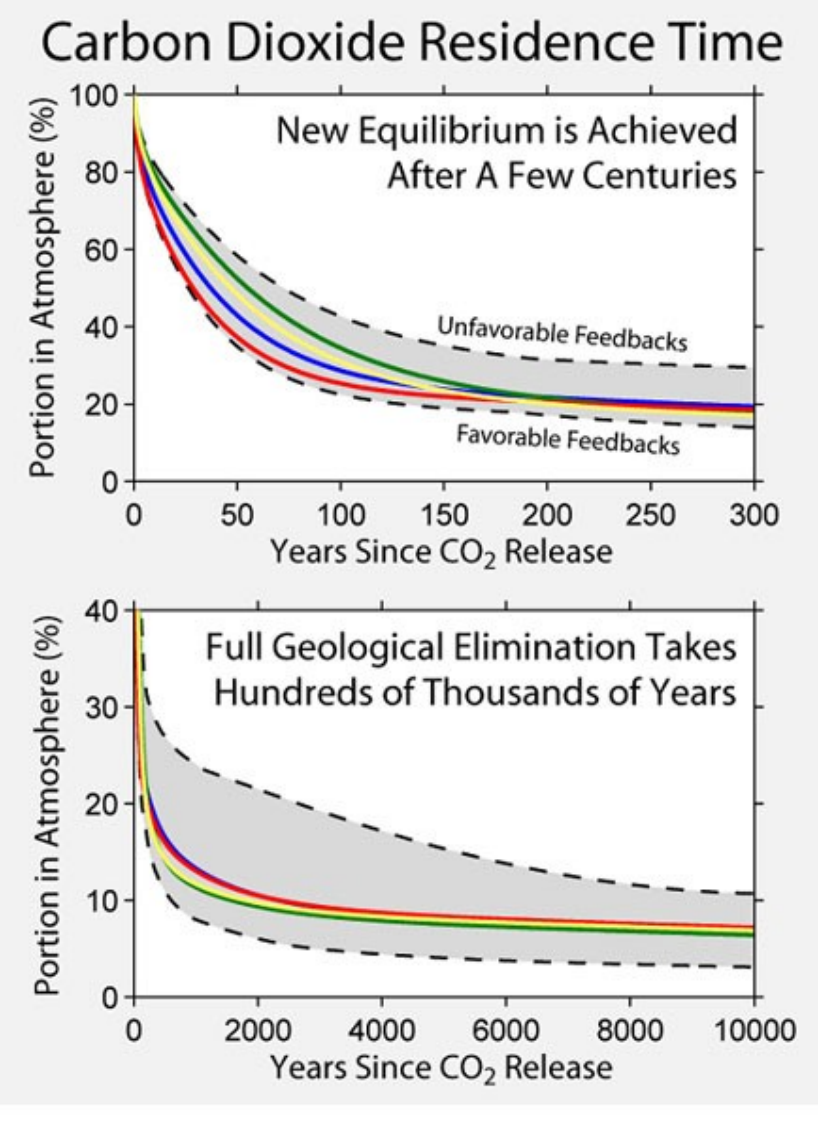
Global Emissions of Carbon Dioxide: 40 Gigatonnes/year

Atmospheric Carbon Dioxide



The ocean reservoir is currently absorbing CO_2

Atmospheric Carbon Dioxide



Depending on the model, CO_2 residence time in the ocean is centuries

Mineral Resources: Silicates as well as:



Important mineral class for many reasons

- Carbonates: 1/5 of world's sedimentary rocks
- Oxides: Major ores
- Sulfides: Major ores
- Unique mechanical and physical properties

Elements in your telephone



Mineral Resources

Element	How used
Aluminum	metal alloy in dial, transmitter, and receiver
Antimony, Arsenic, Beryllium, Bismuth	Alloy in dial
Boron, Germanium, Indium, Silicon	Dial mechanism
Cadmium	Color in yellow plastic housing
Calcium	In lubricant for moving parts
Carbon	Plastic housing, transmitter steel parts
Chlorine	wire insulation
Chromium	Color in green plastic, plating, stainless steel
Cobalt	magnetic material in receiver
Copper	wire, plating, brass piece parts
Fluorine	Plastic piece parts
Gold, Palladium, Platinum	Electrical contacts
Hydrogen	Plastic housing, wire insulation
Iron	Steel, magnetic materials
Krypton	Ringer in Touch-Tone set
Lead, Tin	Solder in connections
Lithium	In lubricant for moving parts
Magnesium	Die castings in transmitter, ringer
Manganese, Phosphorus, Sulfur	Steel in piece parts
Mercury	Color in read plastic housing
Molybdenum	magnet in receiver
Nickel	magnet in receiver, stainless steel parts
Nitrogen	Hardened heat-treated piece parts
Oxygen	Plastic housing, wire insulation
Silver, Tin	Plating
Sodium	In lubricant for moving parts
Tantalum	Integrated circuit in Trimline set
Titanium	Color in white plastic housing
Tungsten	Lights in Princess and key sets
Vanadium	Receiver
Zinc	Brass, die casting in transmitter, ringer

Elements in your telephone

Mineral Resources

USGS
science for a changing world

3 Li lithium 6.94 | 14 Si silicon 28.08 | 19 K potassium 39.09 | 20 C carbon 40.07 | 29 Cu copper 63.54 | 31 Ga gallium 69.72 | 32 Ge germanium 72.63 | 33 As arsenic 74.92 | 47 Ag silver 107.86 | 49 In indium 114.81 | 50 Sn tin 118.71 | 73 Ta tantalum 180.94 | 74 W tungsten 183.84 | 78 Pt platinum 195.08

A World of Minerals in Your Mobile Device

Mobile phones and other high-technology communications devices could not exist without mineral commodities. More than one-half of all components in a mobile device—including its electronics, display, battery, speakers, and more—are made from mined and semi-processed materials (mineral commodities). Some mineral commodities can be recovered as byproducts during the production and processing of other commodities. As an example, bauxite is mined for its aluminum content, but gallium is recovered during the aluminum production process. The images below show the **ore minerals** (sources) of some mineral commodities that are used to make components of a mobile device. On the reverse side, the map and table depict the major source countries producing some of these mineral commodities along with how these commodities are used in mobile devices. For more information on minerals, visit http://minerals.usgs.gov.

Display

A mobile device's glass screen is very durable because glassmakers combine its main ingredient, **silica** (silicon dioxide or quartz) **sand**, with ceramic materials and then add potassium.

Layers of indium-tin-oxide are used to create transparent circuits in the display. Tin is also the ingredient in circuit board solder, and **cassiterite** is a primary source of tin.

Gallium provides light emitting diode (LED) backlighting. **Bauxite** is the primary source of this commodity.

Sphalerite is the source of indium (used in the screen's conductive coating) and germanium (used in displays and LEDs).

Electronics and Circuitry

The content of copper in a mobile device far exceeds the amount of any other metal. Copper conducts electricity and heat and comes from the source mineral **chalcopyrite**.

Tetrahedrite is a primary source of silver. Silver-based inks on composite boards create electrical pathways through a device.

Silicon, very abundant in the Earth's crust, is produced from the source mineral quartz and is the basis of integrated circuits.

Arsenopyrite is a source of arsenic, which is used in radio frequency and power amplifiers.

Tantalum, from the source mineral **tantalite**, is added to capacitors to regulate voltage and improve the audio quality of a device.

Wolframite is a source of tungsten, which acts as a heat sink and provides the mass for mobile phone vibration.

Battery

Spodumene and subsurface brines are the sources of lithium used in cathodes of lithium-ion batteries.

Graphite is used for the anodes of lithium-ion batteries because of its electrical and thermal conductivity.

Speakers and Vibration

Bastnaesite is a source of rare-earth elements used to produce magnets in speakers, microphones, and vibration motors.

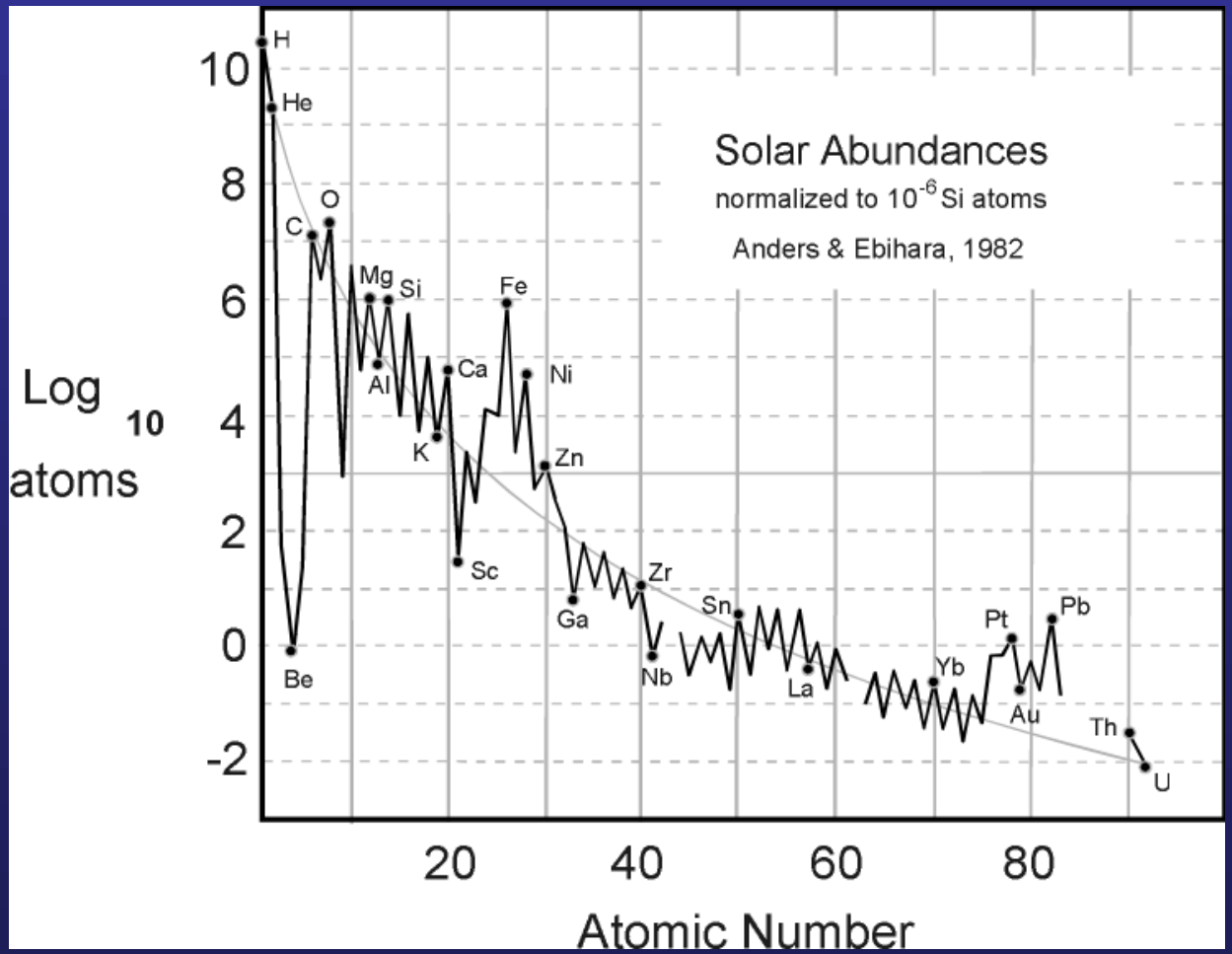


Banner image courtesy of freevector-archive.com

U.S. Department of the Interior
U.S. Geological Survey

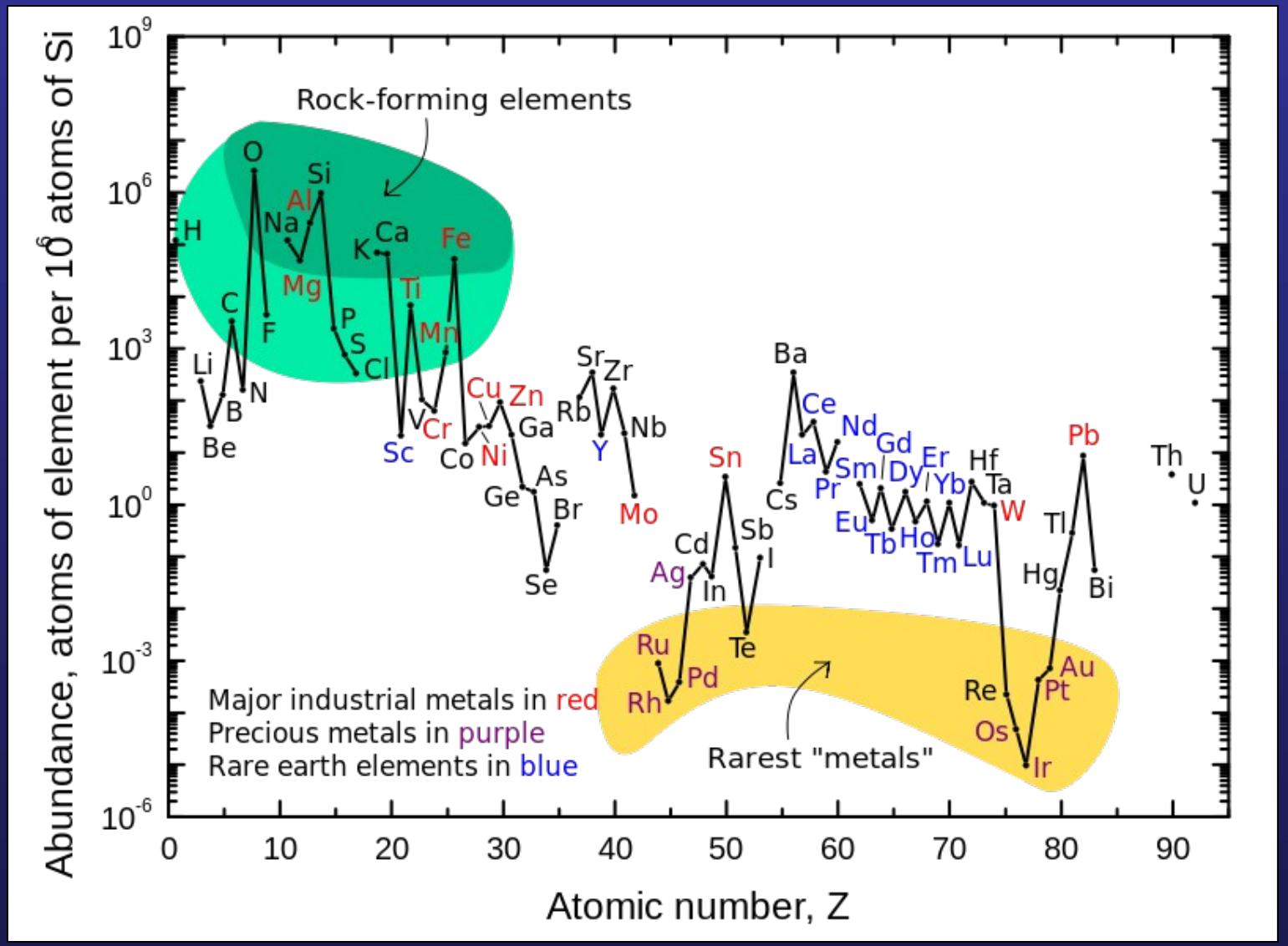
General Information Product 167
September 2016

Elemental Abundances



Note logarithmic y-axis scale

Crustal Elemental Abundances



Most of the Earth's iron, gold, platinum, etc. is in the core

Economic elements and minerals

An ore is a valuable or useful mineral present in “economic” concentrations.

Thus, ore deposits are usually areas where concentrations of minerals of interest are higher than average.

Whether a deposit is economic or not depends on the market value of the mineral of interest, extraction and transformation costs, etc.

Crustal Abundance of Economically Important Elements

NAME	CHEMICAL SYMBOL	ATOMIC NUMBER	CRUSTAL ABUNDANCE (% BY ABUNDANCE)
Aluminum	Al	13	8.00
Iron	Fe	26	5.8
Magnesium	Mg	12	2.77
Potassium	K	19	1.68
Titanium	Ti	22	0.86
Hydrogen	H	1	0.14
Phosphorus	P	15	0.101
Manganese	Mn	25	0.100
Fluorine	F	9	0.460
Sulfur	S	16	0.030
Chlorine	Cl	17	0.019
Vanadium	V	23	0.017
Chromium	Cr	24	0.0096
Zinc	Zn	30	0.0082
Nickel	Ni	28	0.0072
Copper	Cu	29	0.0058
Cobalt	Co	27	0.0028
Lead	Pb	82	0.00010
Boron	B	5	0.0007
Beryllium	Be	4	0.00020
Arsenic	As	33	0.00020
Tin	Sn	50	0.00015
Molybdenum	Mb	42	0.00012
Uranium	U	92	0.00016
Tungsten	W	74	0.00010
Silver	Ag	47	0.000008
Mercury	Hg	80	0.000002
Platinum	Pt	78	0.0000005
Gold	Au	79	0.0000002

Concentration Factors

Table 22.1 Economical Concentration Factors of Some Commercially Important Elements

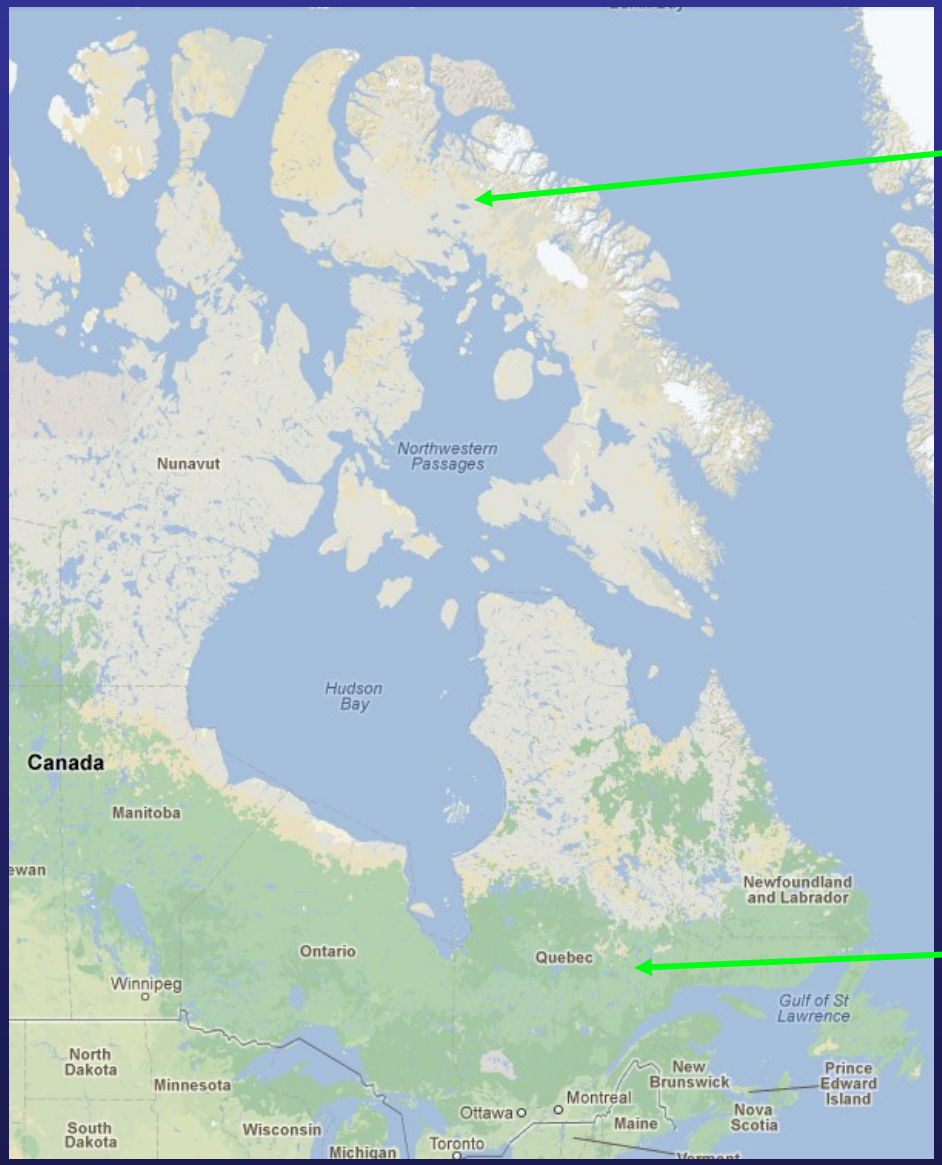
Element	Crustal Abundance (Percent by Weight)	Economical Concentration Factor[1]
Aluminum	8.00	3–4
Iron	5.8	5–10
Copper	0.0058	80–100
Nickel	0.0072	150
Zinc	0.0082	300
Uranium	0.00016	1,200
Lead	0.00010	2,000
Gold	0.0000002	4,000
Mercury	0.000002	100,000

[1]Concentration factor = abundance in deposit divided by crustal abundance.

SOURCES: Data from B. J. Skinner, *Earth Resources,* Prentice Hall, 1969; and D. A. Brobst and W. P. Pratt, *Mineral Resources of the U.S.,* USGS Professional Paper 820, 1973.

Depends on crustal abundance and cost of extraction

Concentration Factors: Iron



St. Mary River (Baffin Island) mine:
65% iron oxide
Concentration factor of 10-12

Labrador Trough:
20-30% iron oxide
Concentration factor of 4-5

St Mary River, Baffin

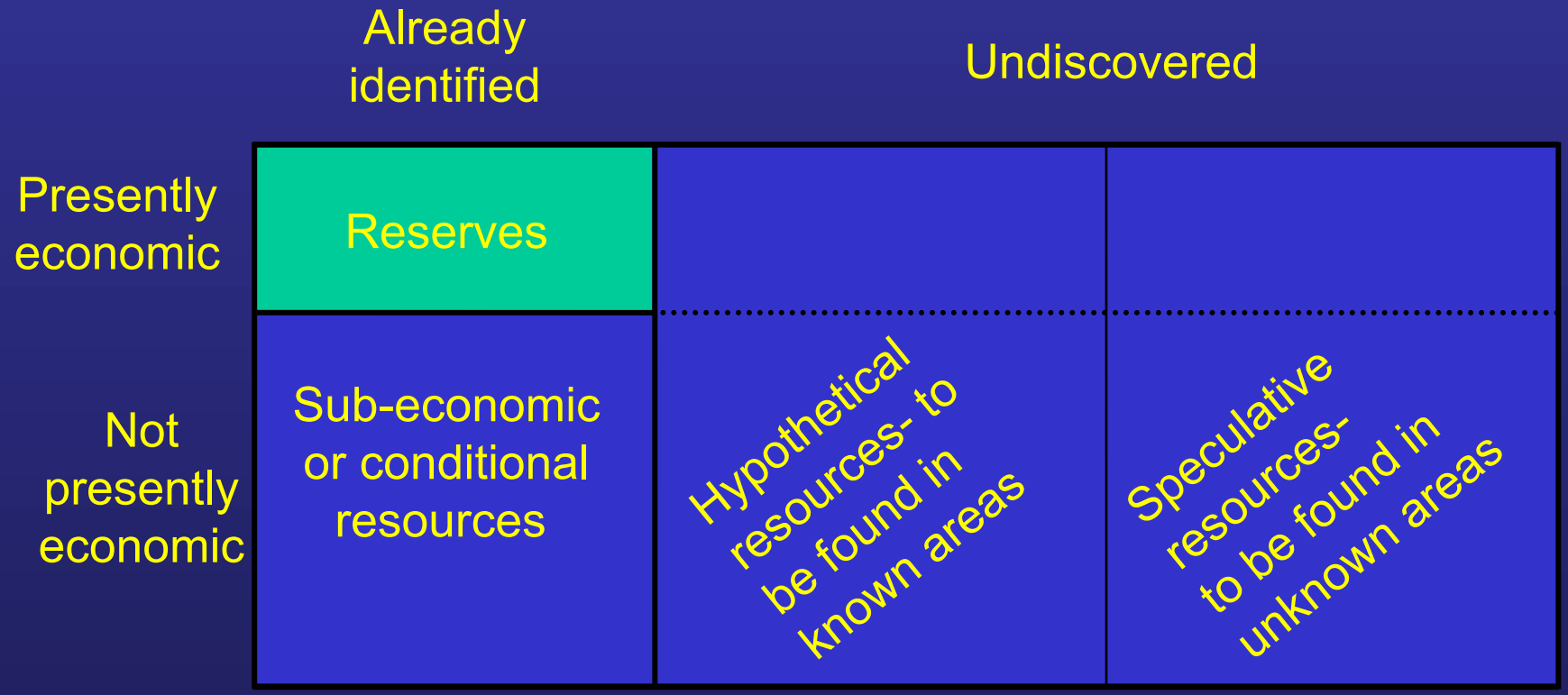


Need to construct everything:
Railway
Port
Ice-breaking freighters

Located on Permafrost: if it melts, it's a swamp



Reserves and Resources



In order to assess how much of a given mineral is available and will be available in the future, estimates based on volumes and concentrations of the minerals in known deposits, and predictions of the future are made.

Concentrating Minerals and Metals

Materials must be concentrated by some natural process before they are economic to mine.

- Hydrothermal: hot, water-rich solutions
  - Gold in Quartz veins
- Metamorphic: recrystallization due to heat and pressure
  - The minerals Corundum or Garnet or Diamond
- Magmatic: fractional crystallization or melting
  - Platinum, Copper, other base metals
- Sedimentary: precipitation from a lake or seawater
  - Salt, Gypsum, Borax, Barium
- Placer: the action of waves or currents
  - Gold in gravel bars
- Residual: weathering processes
  - Bauxite (Aluminum ore)

California: Hydraulic Mining





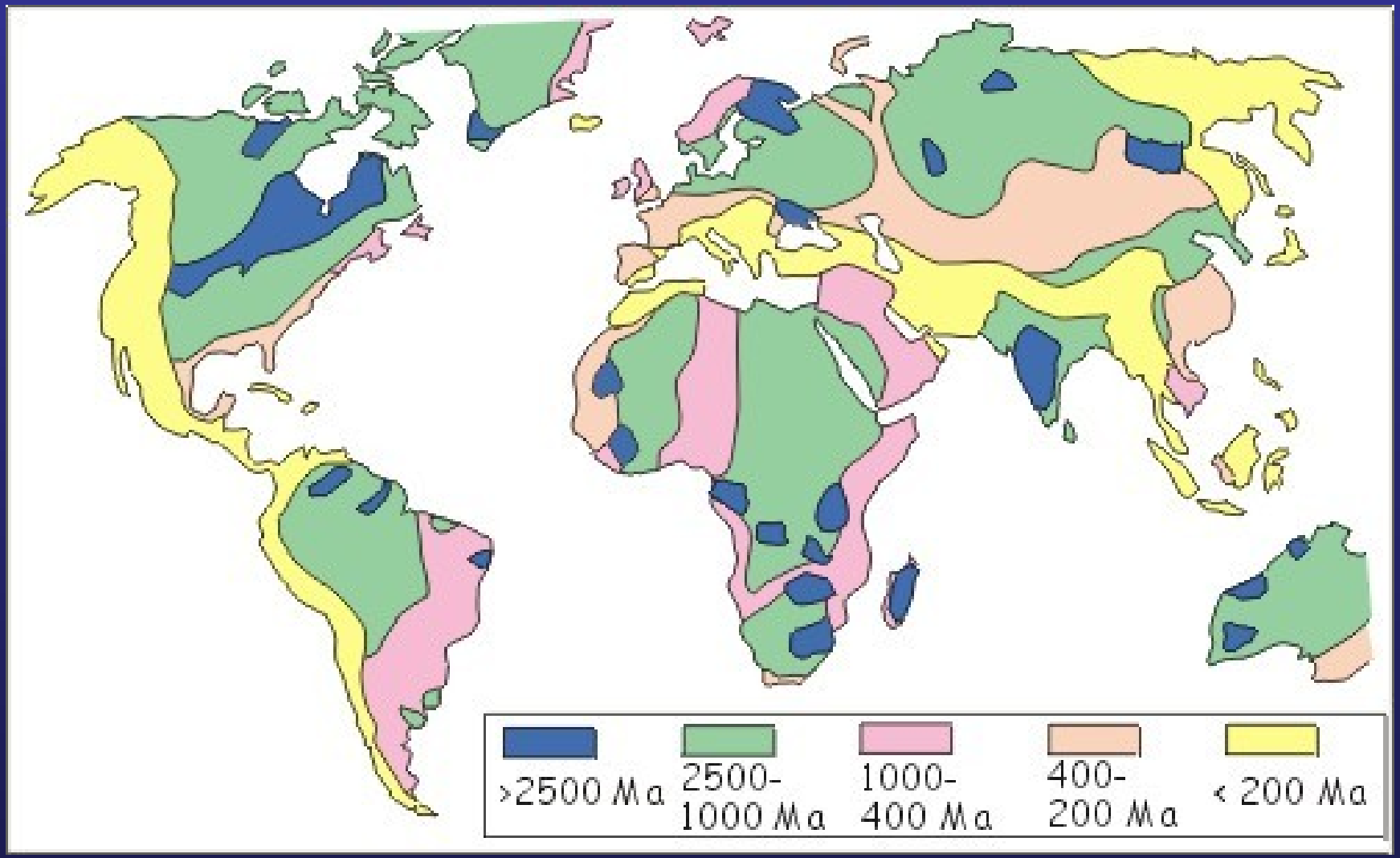
Hydraulic Mining

Las Médulas, NW Spain



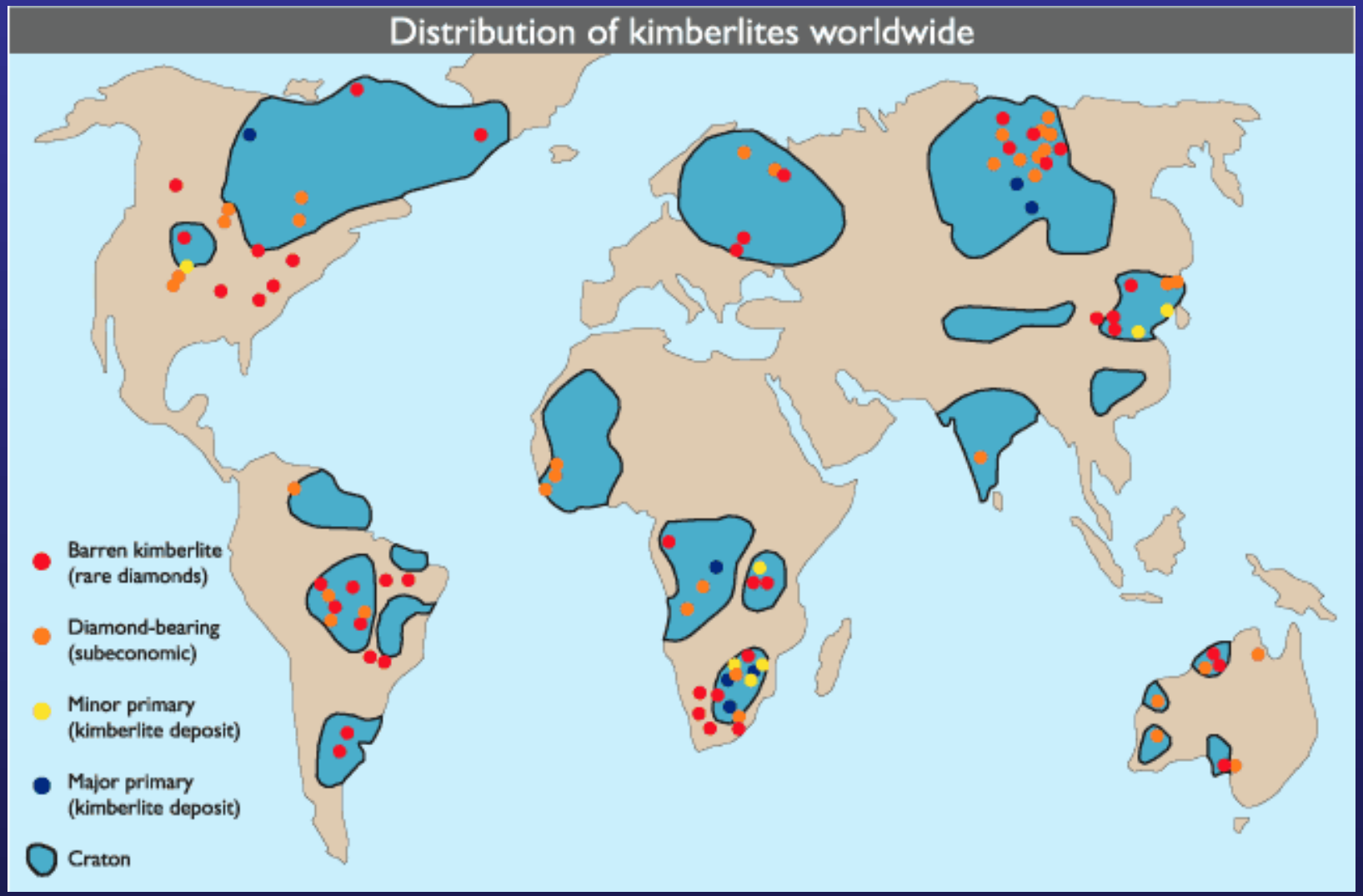
Roman Hydraulic Mining

Plate Tectonics: Continental Crust Age



Ma = millions of years

Continental Crust Age: Diamonds



Kimberlites are the igneous rock where diamonds are found

Magmatic Deposits: Diamonds



(a)

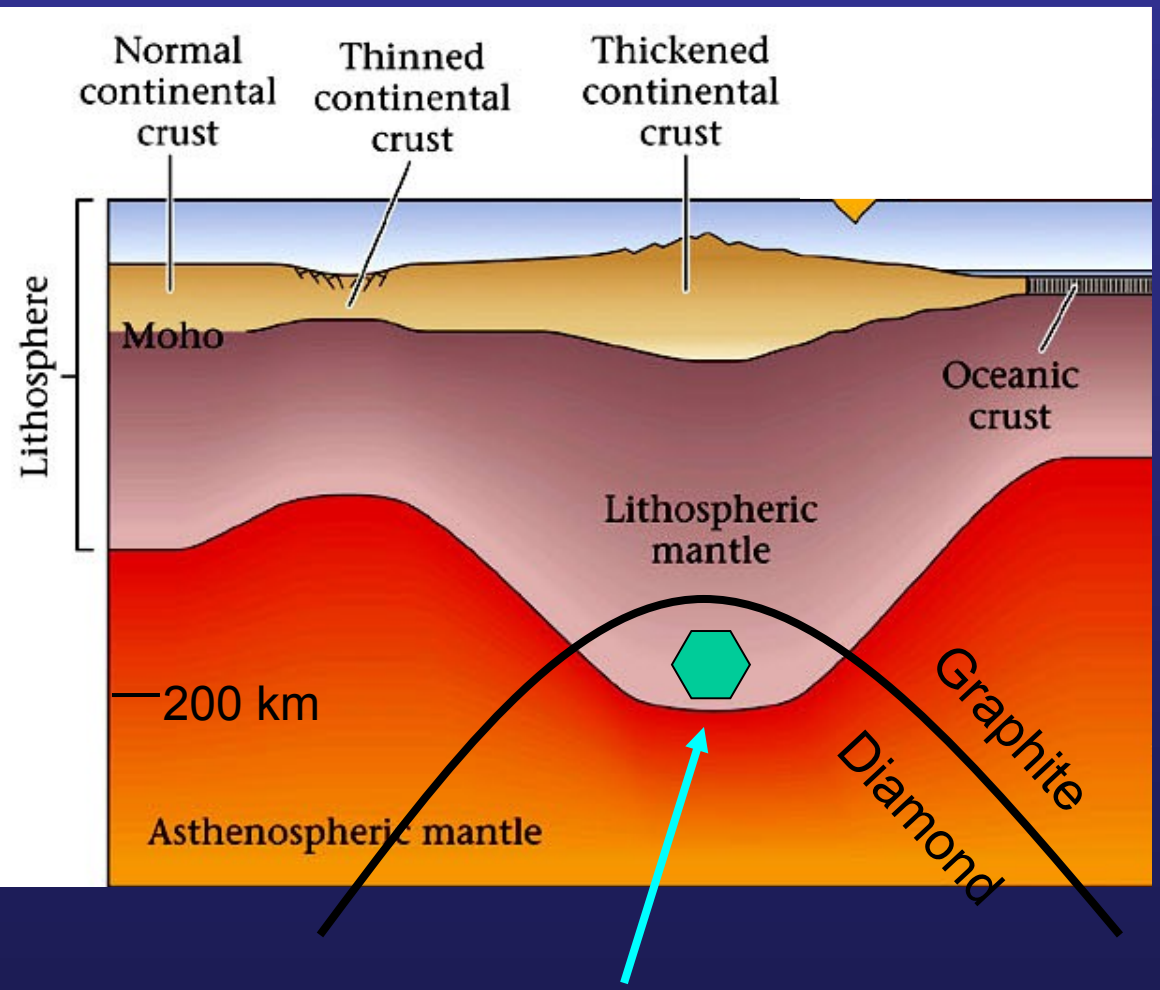
The ‘Big Hole’ at Kimberly, South Africa

Kimberlite rock containing diamond



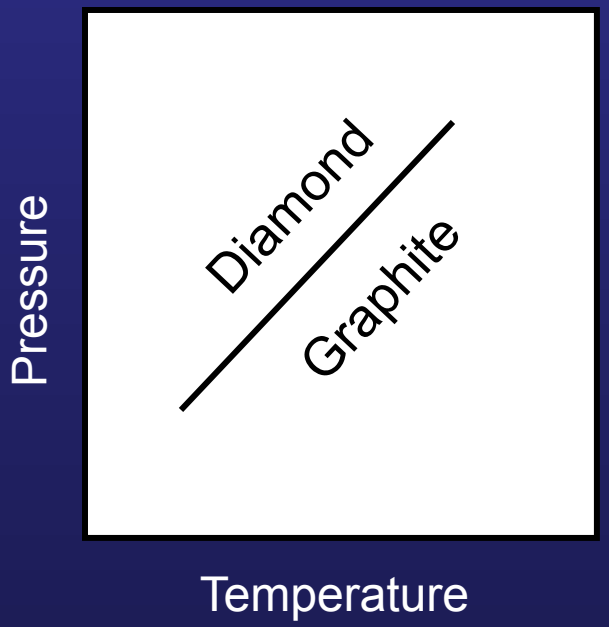
(b)

Diamond formation conditions

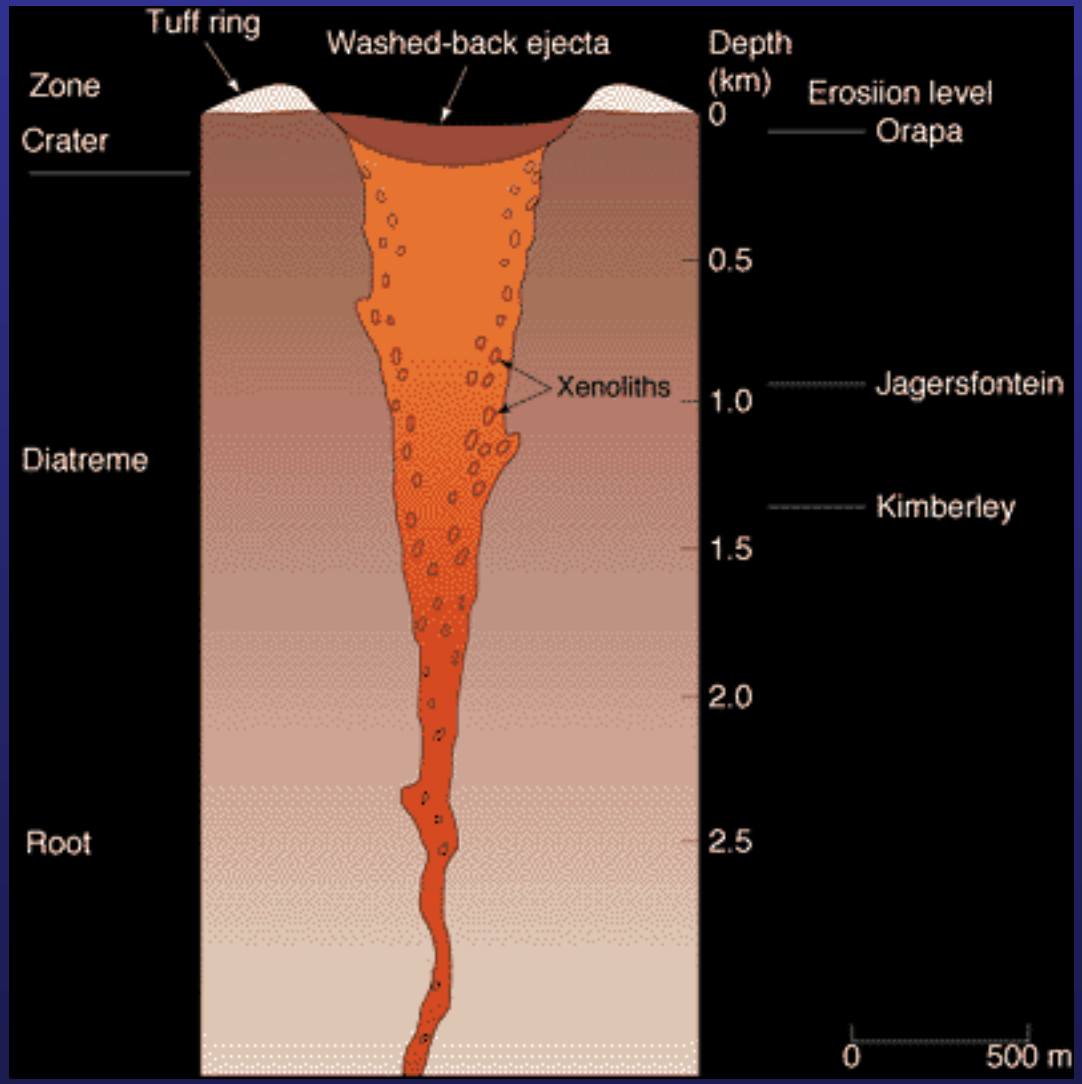


Diamond formation area

The carbon phase diagram



Kimberlite Pipe



A supersonic eruption of magma and gas brings the diamonds to the surface before they transform back to graphite

2019 World Diamond Production

Producing Country	Carats (000)	Ave price/ carat US$	Value US $Billion
Russia	45,271	91	4.12
Botswana	23,687	145	3.43
Australia	12,999	12	0.16
D.R. Congo	14,158	16	0.23
Canada	18,638	91	1.70
Angola	9,150	138	1.27
South Africa	7,181	122	0.87
Zimbabwe	2,108	67	0.14

The Firefox Diamond



187 carats

Diamonds in the NW Territories



The ice road to the mine.

The Jerrico mine in Nunavut (Canada's third diamond mine) failed, in part, because the winter wasn't cold enough to support the ice road for more than one month

Diamonds in the NW Territories



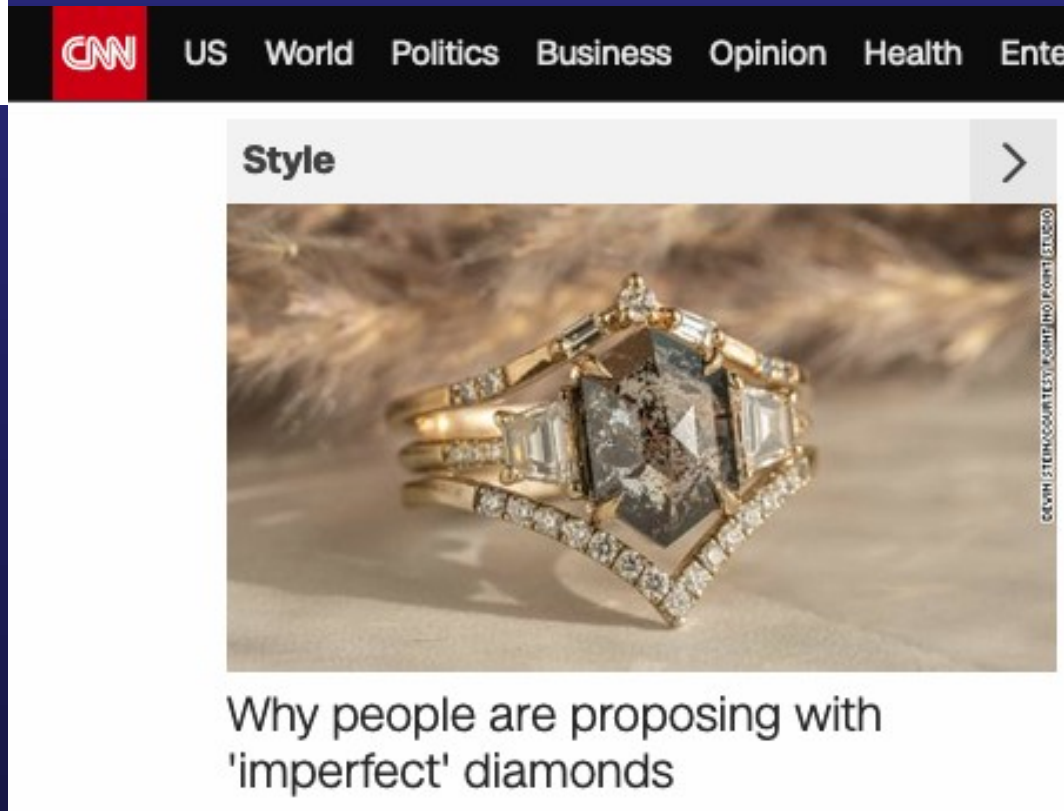
The Diavik Mine, near Yellowknife

Artificial Diamonds, Marketing



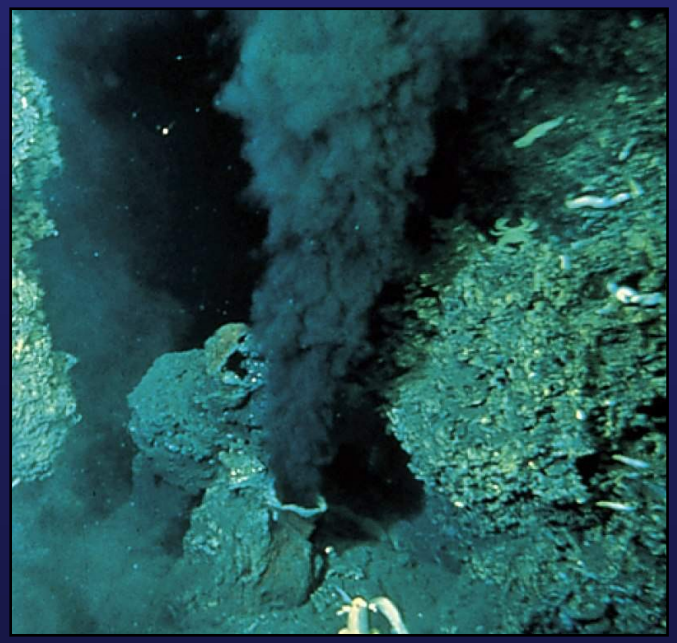
Diamonds can now be commercially produced for less than they can be mined…

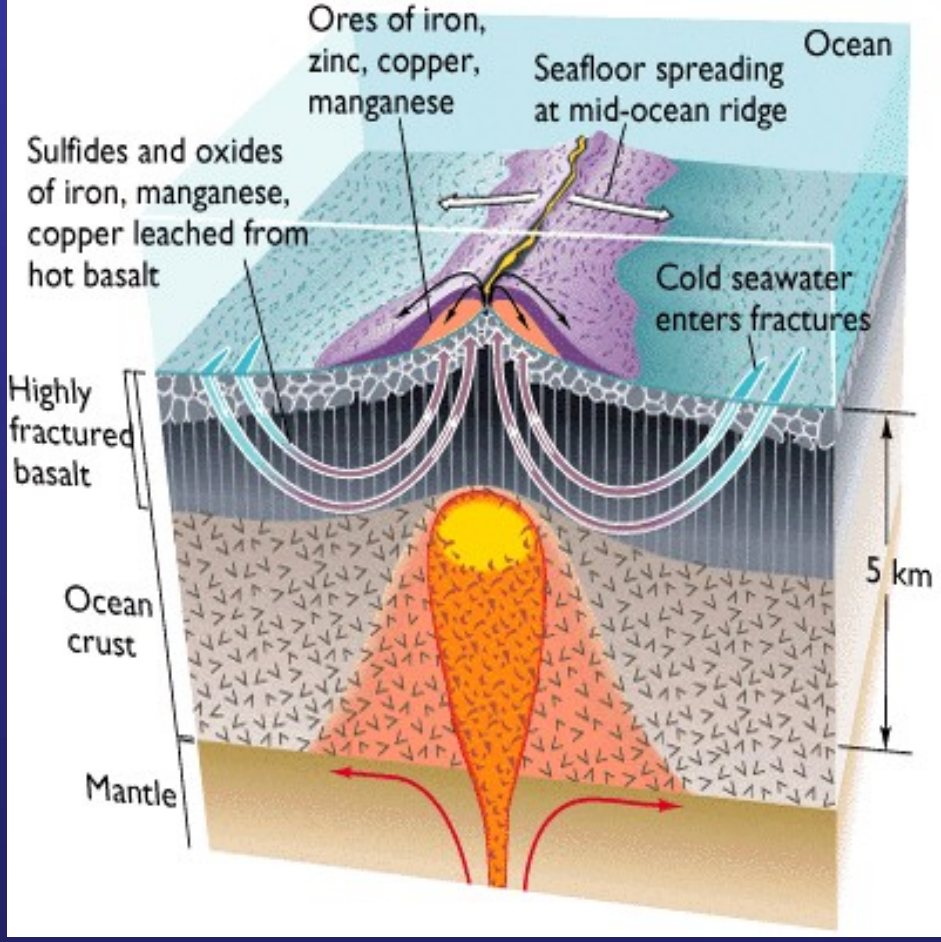
So, diamond miners are now marketing ‘imperfect’ natural diamonds as ‘better’.



Hydrothermal Deposits

Hydrothermal deposits form when hot magma heats groundwater in adjacent rocks. The hot water leaches metals from the rock. When the water cools near or at the surface, these metals precipitate, forming deposits.





Hydrothermal vents at mid-oceanic ridges (black smokers) deposit metal-rich sulfides, forming large, massive deposits (copper, zinc, etc).

Magmatic/ Hydrothermal Ore Deposits



The Bingham Canyon Mine, Utah

The world's largest human excavation: 4 km (2.5 miles) wide by 0.9 km (0.5 mile) deep. The ore was formed as a magma cooled and expelled copper-rich fluids into the surrounding rock.



Bingham Copper

Dump trucks for scale: each holds 240 tons per load.

The ore contains 0.5% Copper -that's 5 kg of copper for each metric tonne (1000 kg) of ore mined -and a second tonne of overburden removed.

Copper is currently $3.30/lb or $7.25/kg. This means the Bingham mine has to move 2 tonnes of rock, and process 1 tonne of ore for about $36, on average. A large open-pit mine is the only economic method to do this.

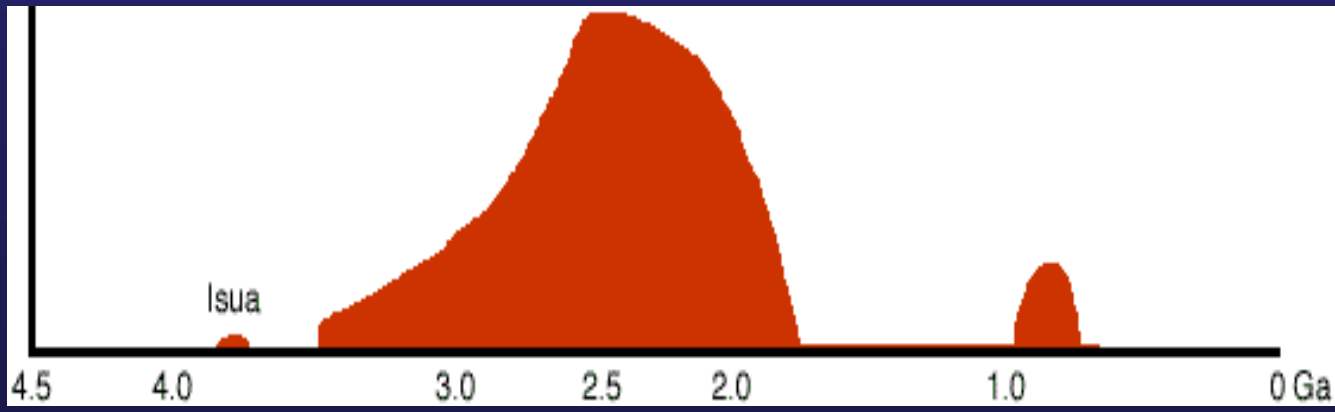
Sedimentary deposits: iron formations





Iron formations, are formed by precipitation of iron oxide from seawater.

Iron formation are restricted in geologic time.



Age of iron formations (in billion years)

Life changing the planet

$CO_2 + H_2O + light \Rightarrow sugars + O_2$

Photosynthesis

Side effects: protection from UV radiation (ozone) and enabling multi-cellular life

$Fe^{2+} + O_2 \Rightarrow Fe^{3+}$

ferrous iron + oxygen => ferric iron

Soluble Insoluble

Side effects: iron formations

nutrient deficiency in the open ocean

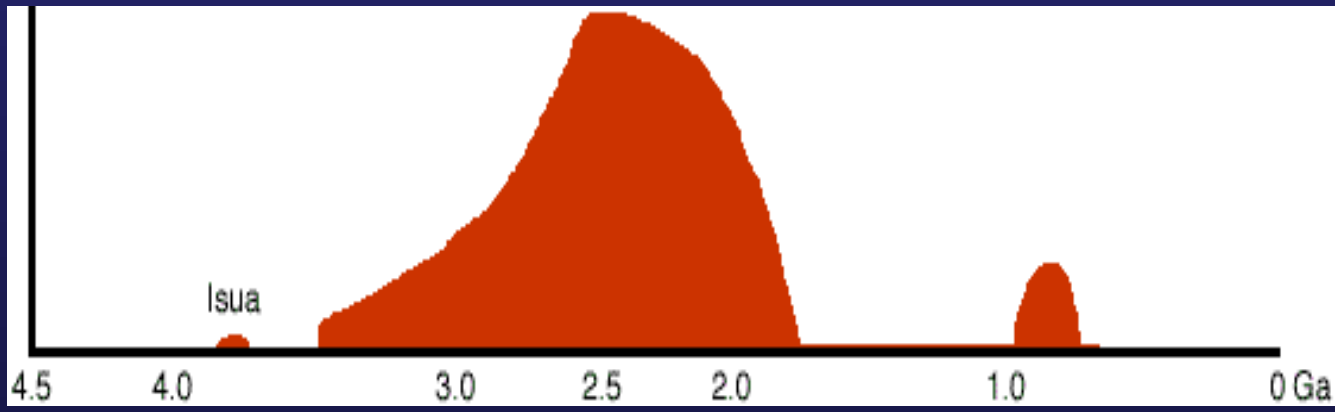
Sedimentary deposits: iron formations





Iron formations, are formed by precipitation of iron oxide from seawater.

Iron formation are restricted in geologic time.



Age of iron formations (in billion years)

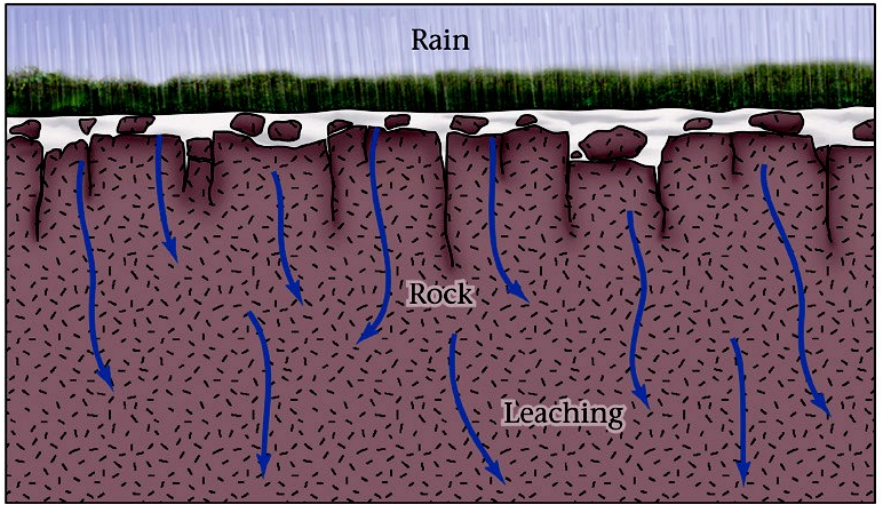
Residual Mineral Deposits

Extreme chemical weathering leaches the soil and regolith

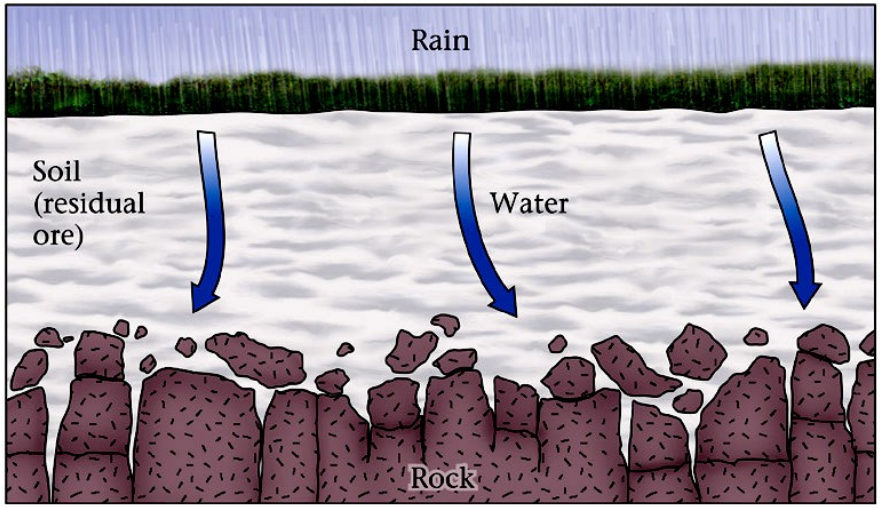
(heat and water required)

Bauxite is what is left

(aluminum oxide ore)



(a)



(b)

Earth: Portrait of a P

Copyright (c) W.W. No

It takes a large amount of power to convert aluminum oxide ore to aluminum metal; Quebec has cheap hydropower…

Atmosphere and Ocean Circulation

Supplemental Readings for:

Lectures 5, 6: Atmosphere energy and circulation:

Kump, Kasting, & Crane, Chapter 3

and

Ahrens' textbook, Chapter 2

Lectures 7, 8: Ocean Circulation:

Kump, Kasting, & Crane, Chapters 4-5

PDFs are in the ‘Readings’ section of myCourses, with the Kump chapters labelled ‘The Earth System’

YaleEnvironment360

Published at the Yale School of Forestry & Environmental Studies

Phosphate: A Critical Resource Misused and Now Running Low

Phosphate has been essential to feeding the world since the Green Revolution, but its excessive use as a fertilizer has led to widespread pollution and eutrophication. Now, many of the world's remaining reserves are starting to be depleted.

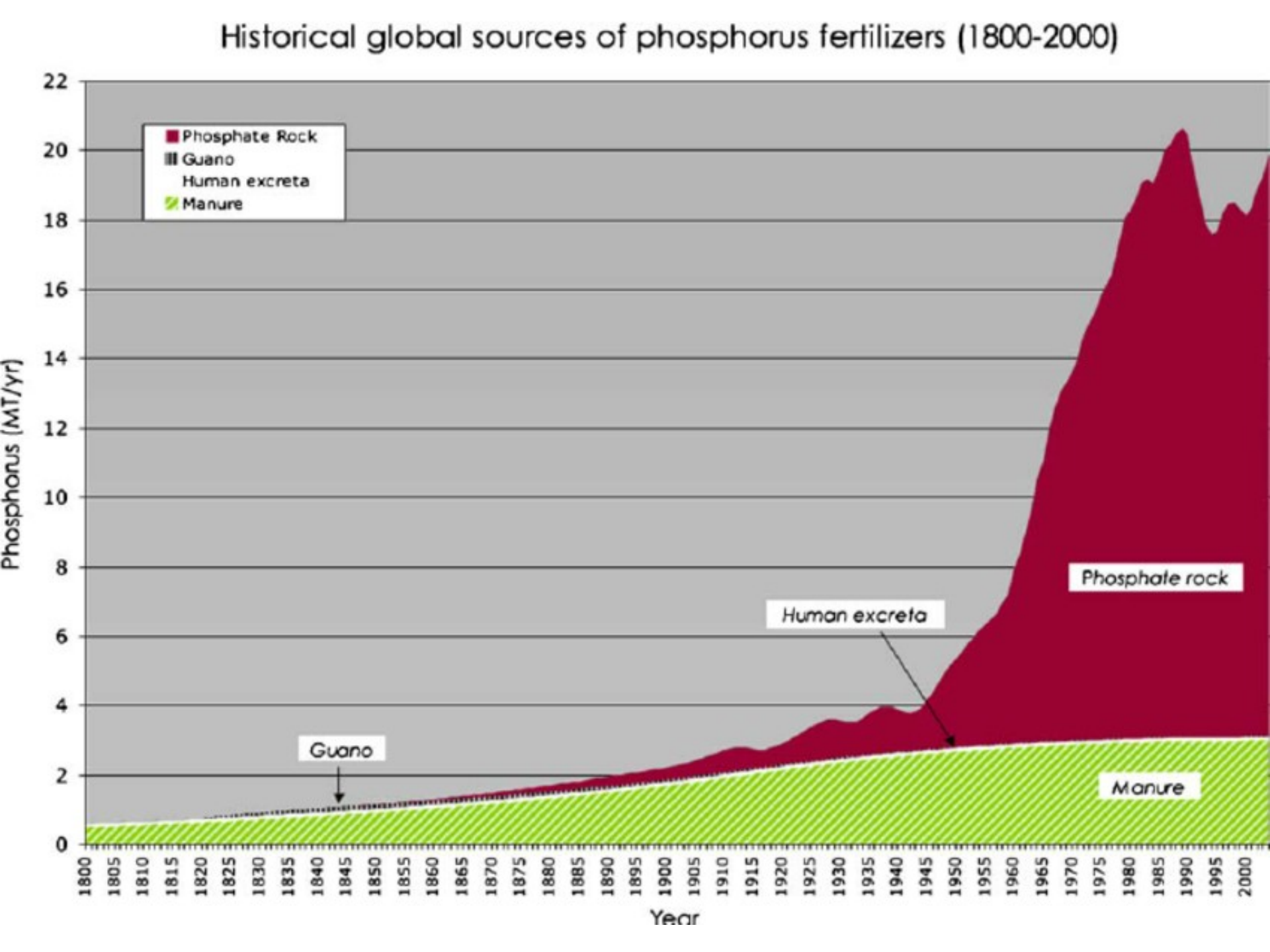
BY FRED PEARCE · JULY 7, 2011

Dr Dana Cordell, *Research Principal*, Institute for Sustainable Futures, University of Technology Sydney (UTS) Australia

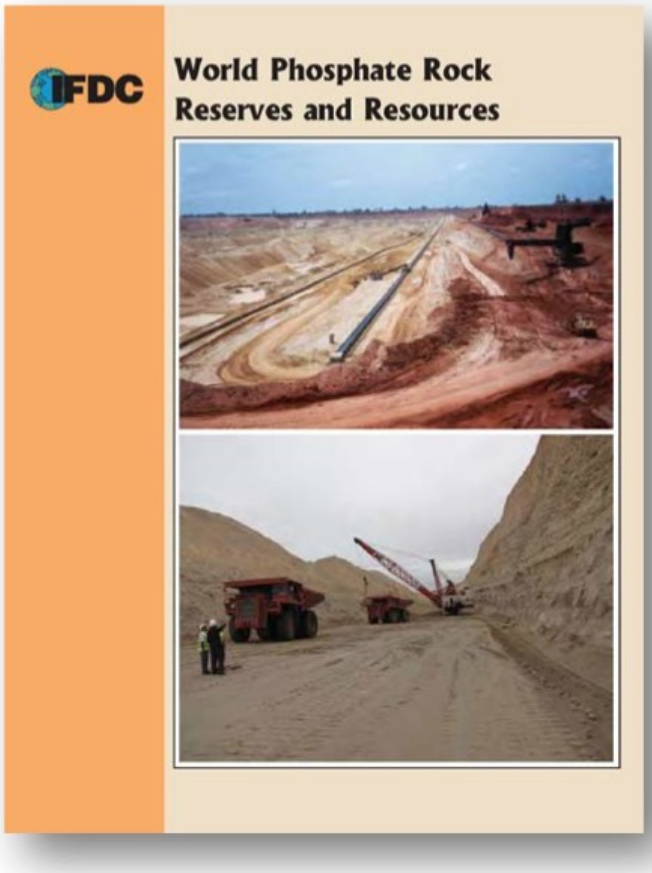
Read the full article: **Life's Bottleneck: Sustaining the World's Phosphorus for a Food Secure Future**

1. Phosphorus equals food

Without phosphorus, we can't produce food. Phosphorus is as essential as water, carbon or oxygen.


Historical global sources of phosphorus fertilizers (1800-2000)
Phosphate Rock
Guano
Human excreta
Manure
Phosphate rock
Human excreta
Guano
Manure
Phosphorus (MT/yr)
Year
0
2
4
6
8
10
12
14
16
18
20
22
1800
1805
1810
1815
1820
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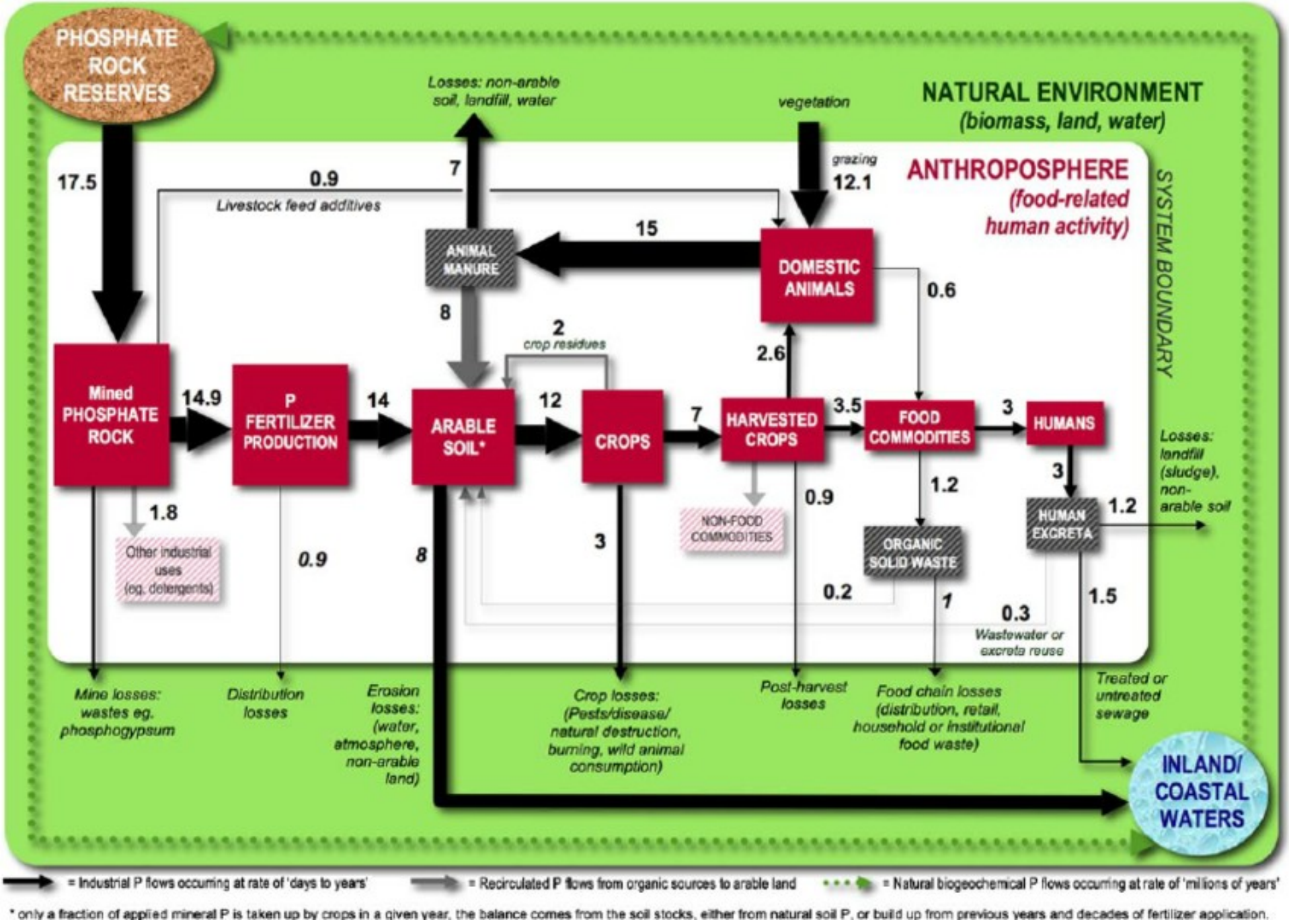
Phosphorus Reserves



Country	IFDC Reserves[a] (Product)	IFDC Resources[b]
	(mmt)	
United States	1,800	49,000
Australia	82	3,500
Brazil	400	2,800
Canada	5	130
China	3,700	16,800
Egypt	51	3,400
Israel	220	1,600
Jordan	900	1,800
Morocco	51,000	170,000[c]
Russia	500	4,300
Senegal	50	250
South Africa	230	7,700
Syria	250	2,000
Togo	34	1,000
Tunisia	85	1,200
Other countries	600[d]	22,000[e]
World total (rounded)	60,000	290,000

a. Reserves as usable or marketable product.
b. Resources as unprocessed phosphate rock of varying grades or concentrate.
c. Including hypothetical resources based on the area limits of the deposits, Morocco resources may be about 340,000 mmt.

But: Morocco is a dictatorship that represses its population…



* only a fraction of applied mineral P is taken up by crops in a given year, the balance comes from the soil stocks, either from natural soil P, or build up from previous years and decades of fertilizer application.

Reclaiming Phosphorus



(a) A urine-diverting dry squat toilet from India (photo: S.Vishwanath)
(b) a urine-diverting dry toilet from Sweden (photo: Dana Cordell)